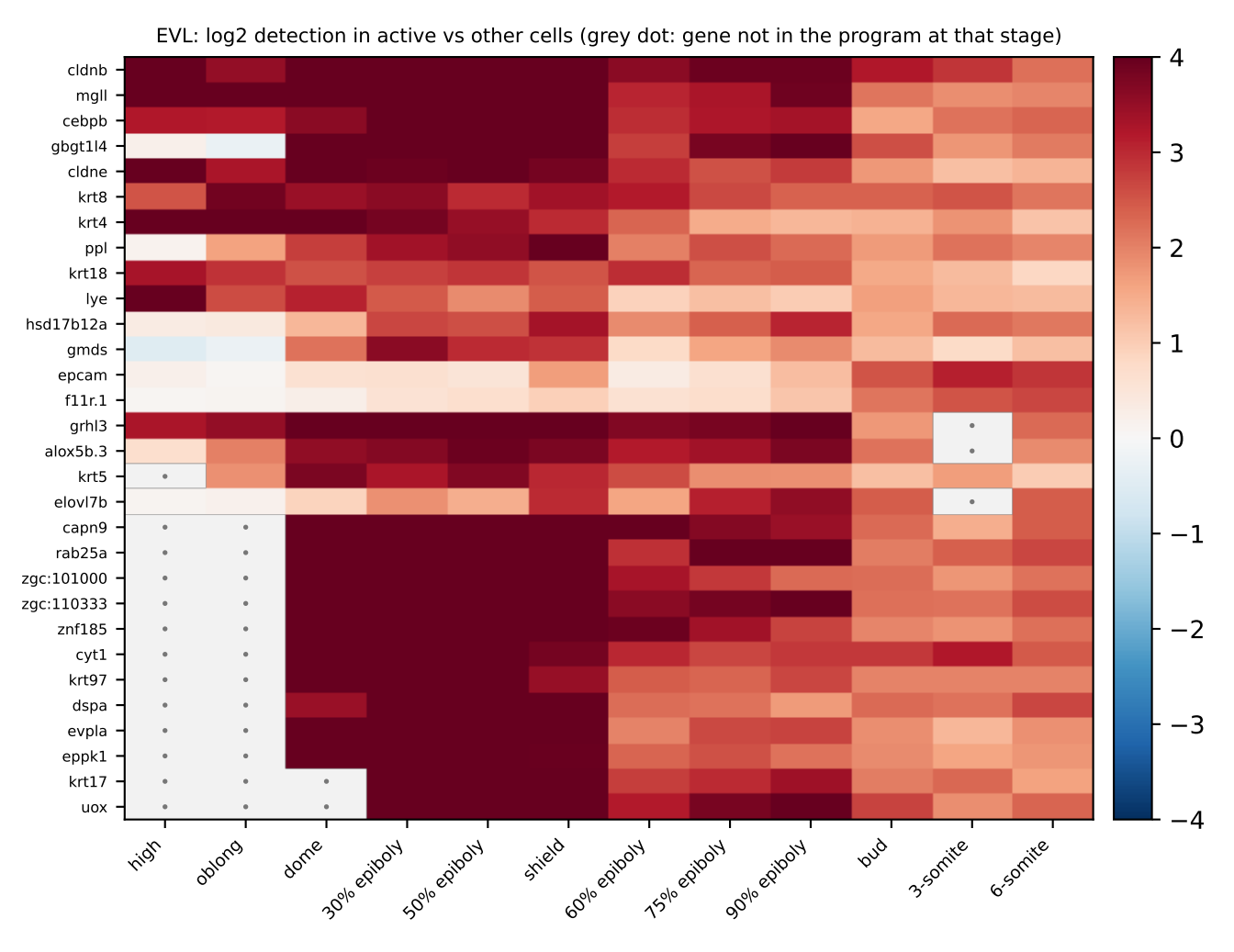
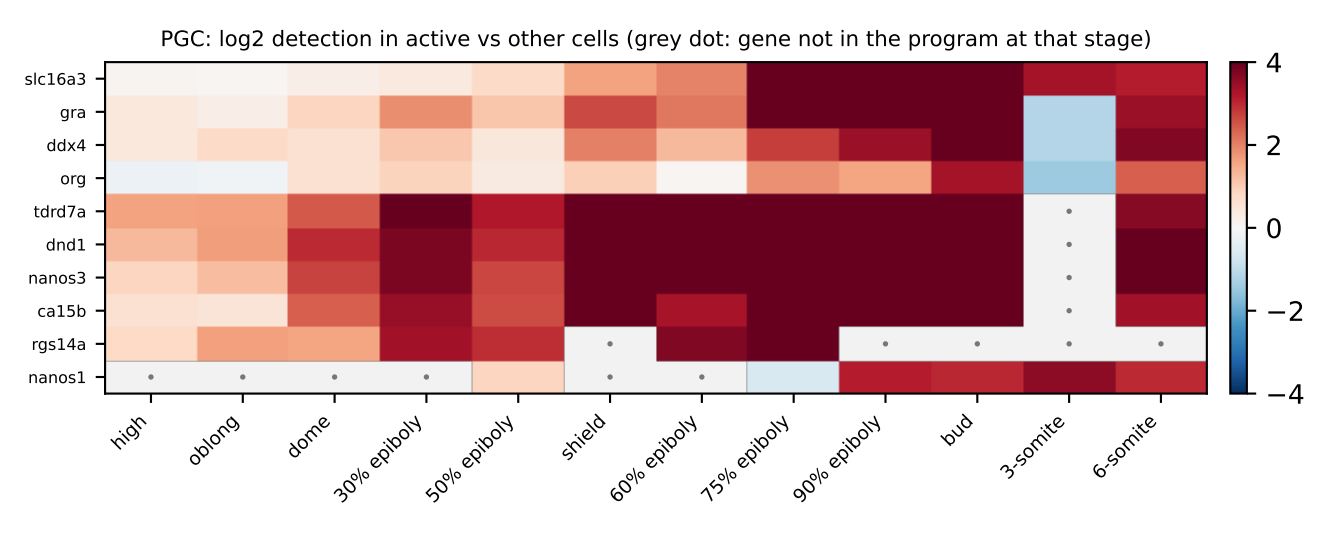
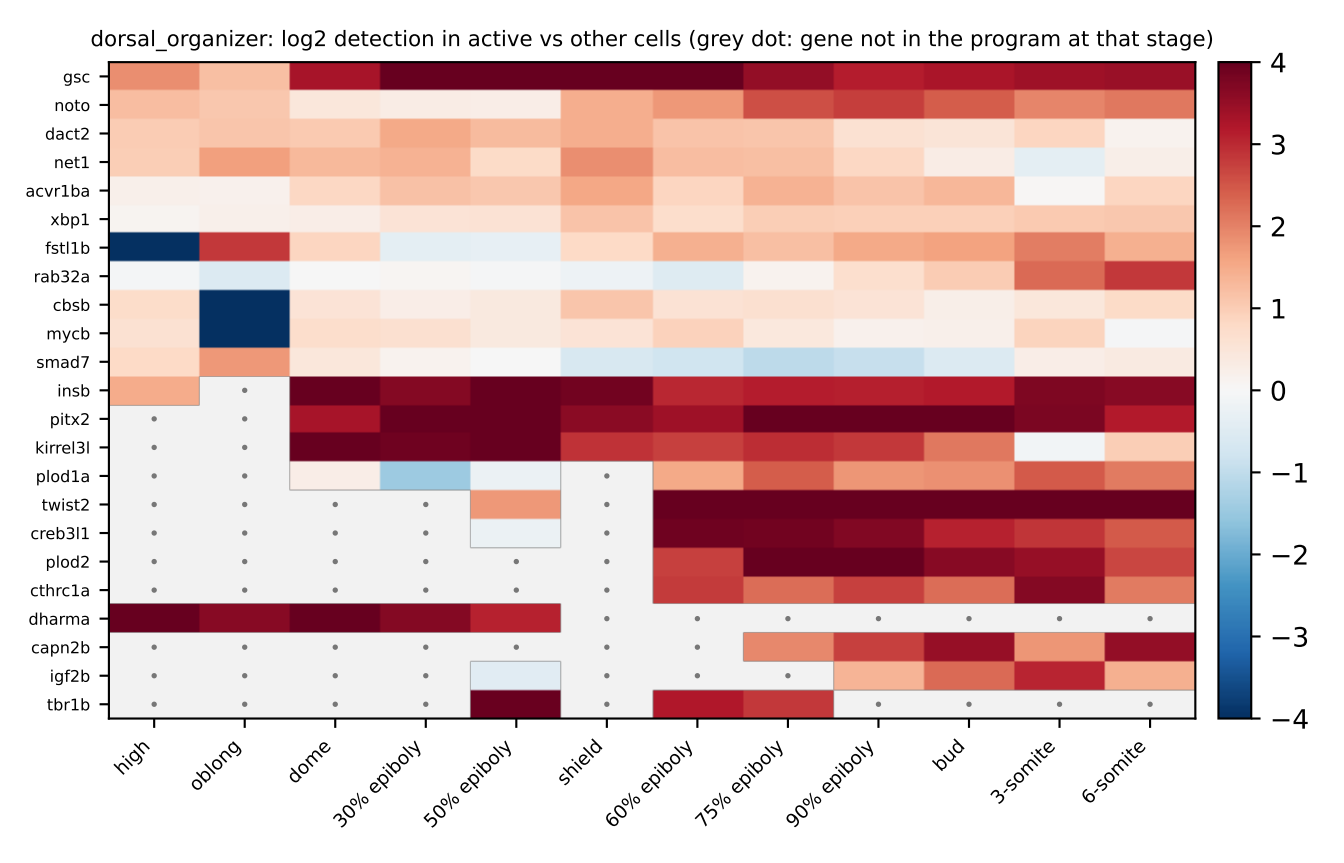
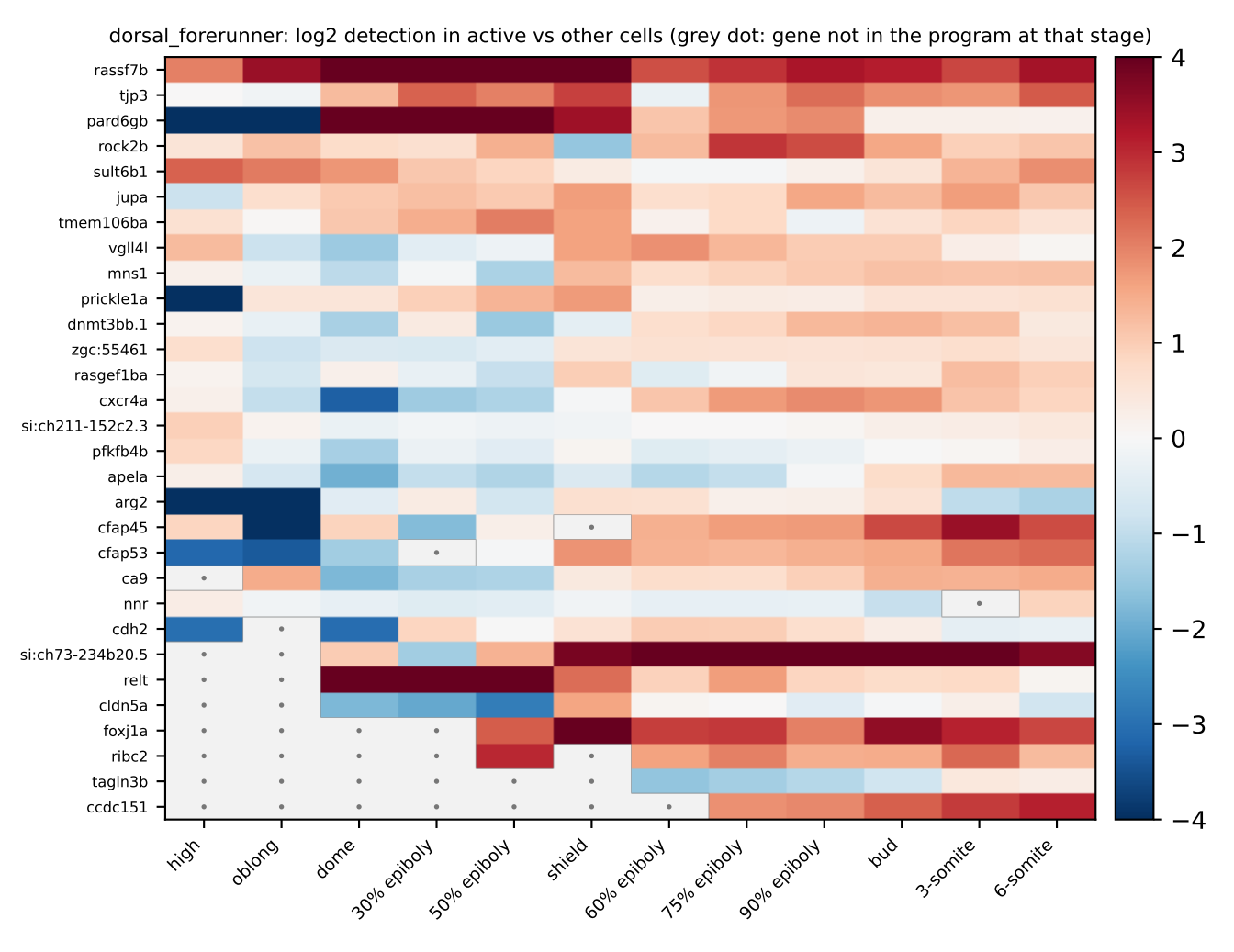
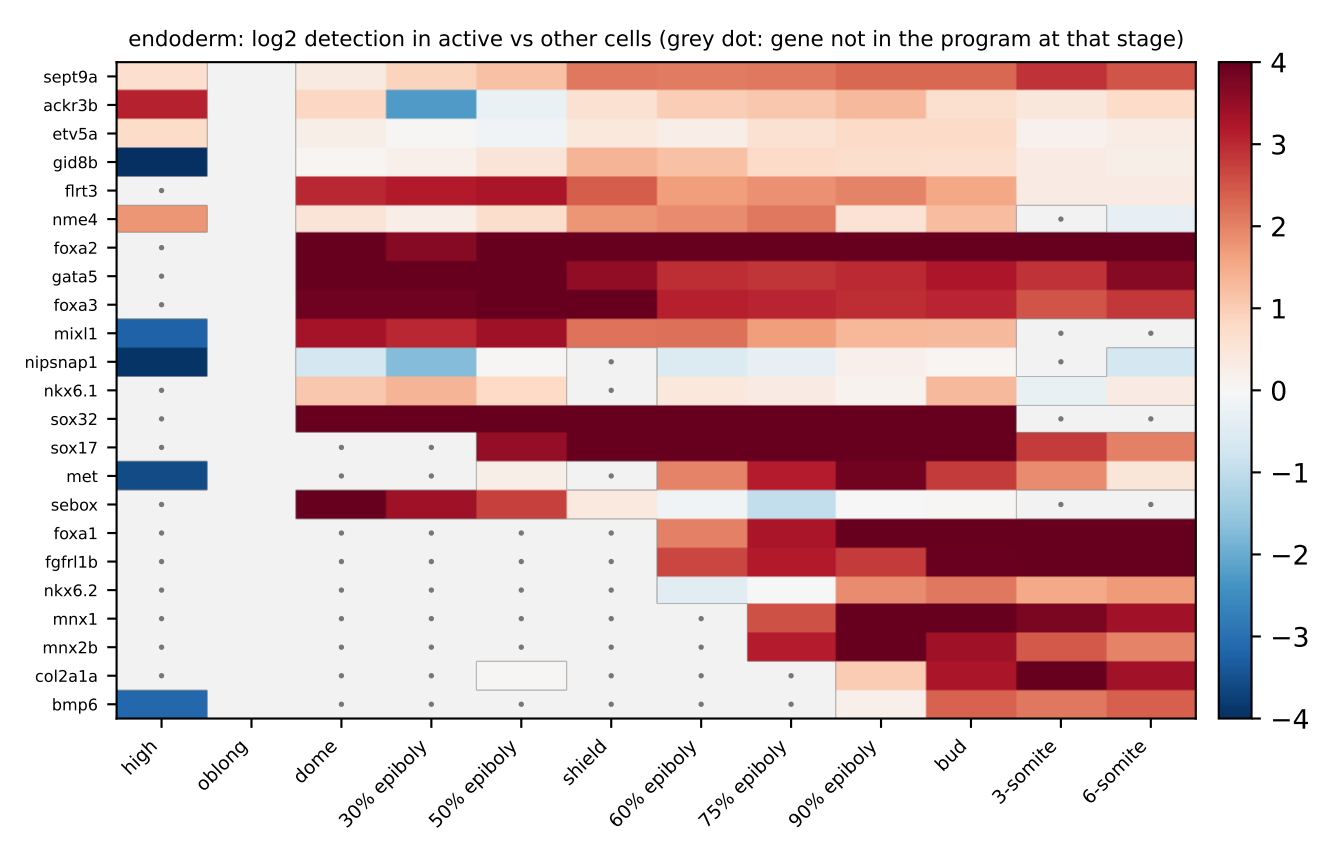
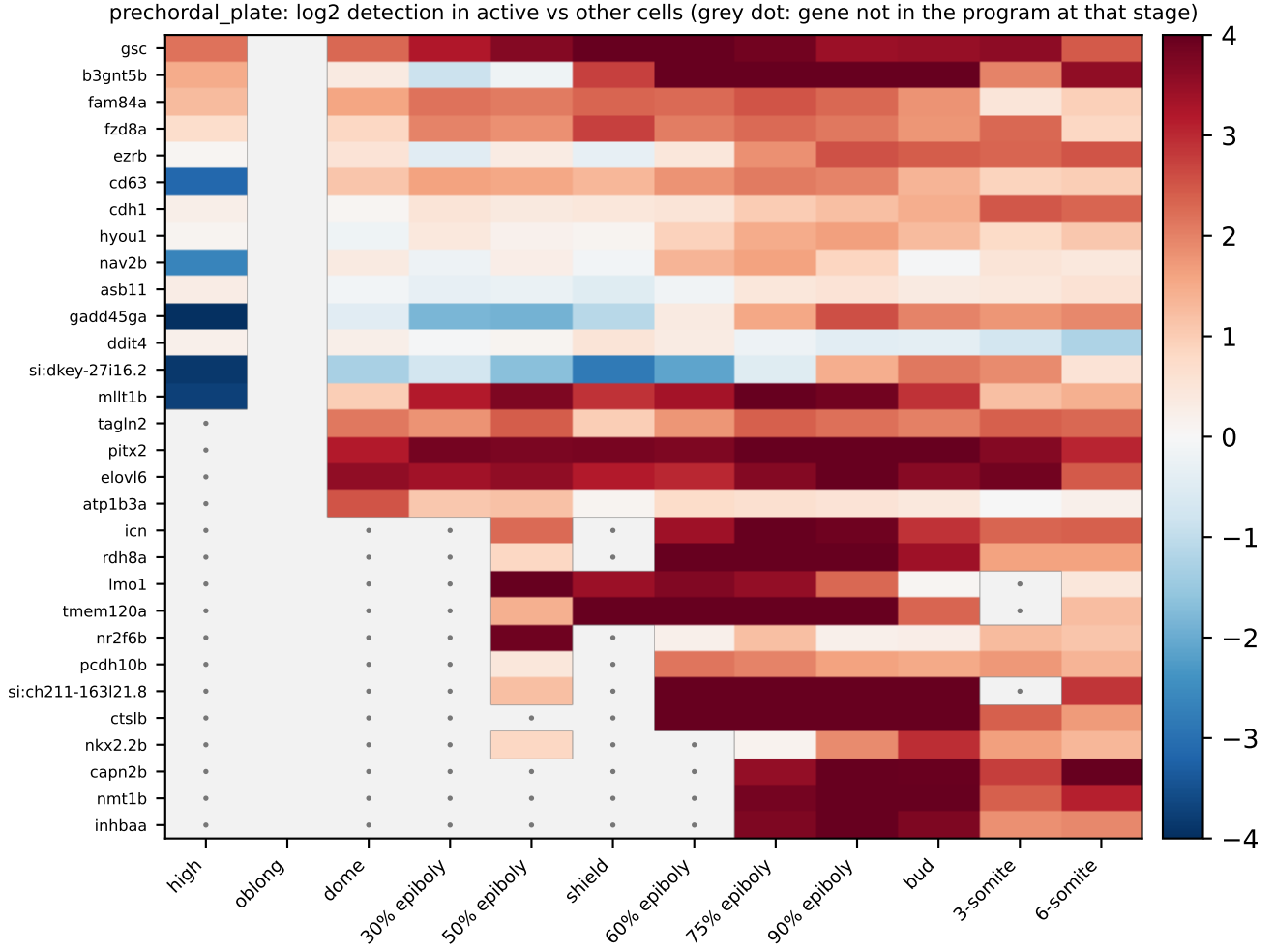

EVL: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
cldnb
mgll
cebpb
gbgt1l4
cldne
krt8
krt4
ppl
krt18
lye
hsd17b12a
gmds
epcam
f11r.1
grhl3
alox5b.3
krt5
elovl7b
capn9
rab25a
zgc:101000
zgc:110333
znf185
cyt1
krt97
dspa
evpla
eppk1
krt17
uox
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
3
2
1
0
−1
−2
−3
−4

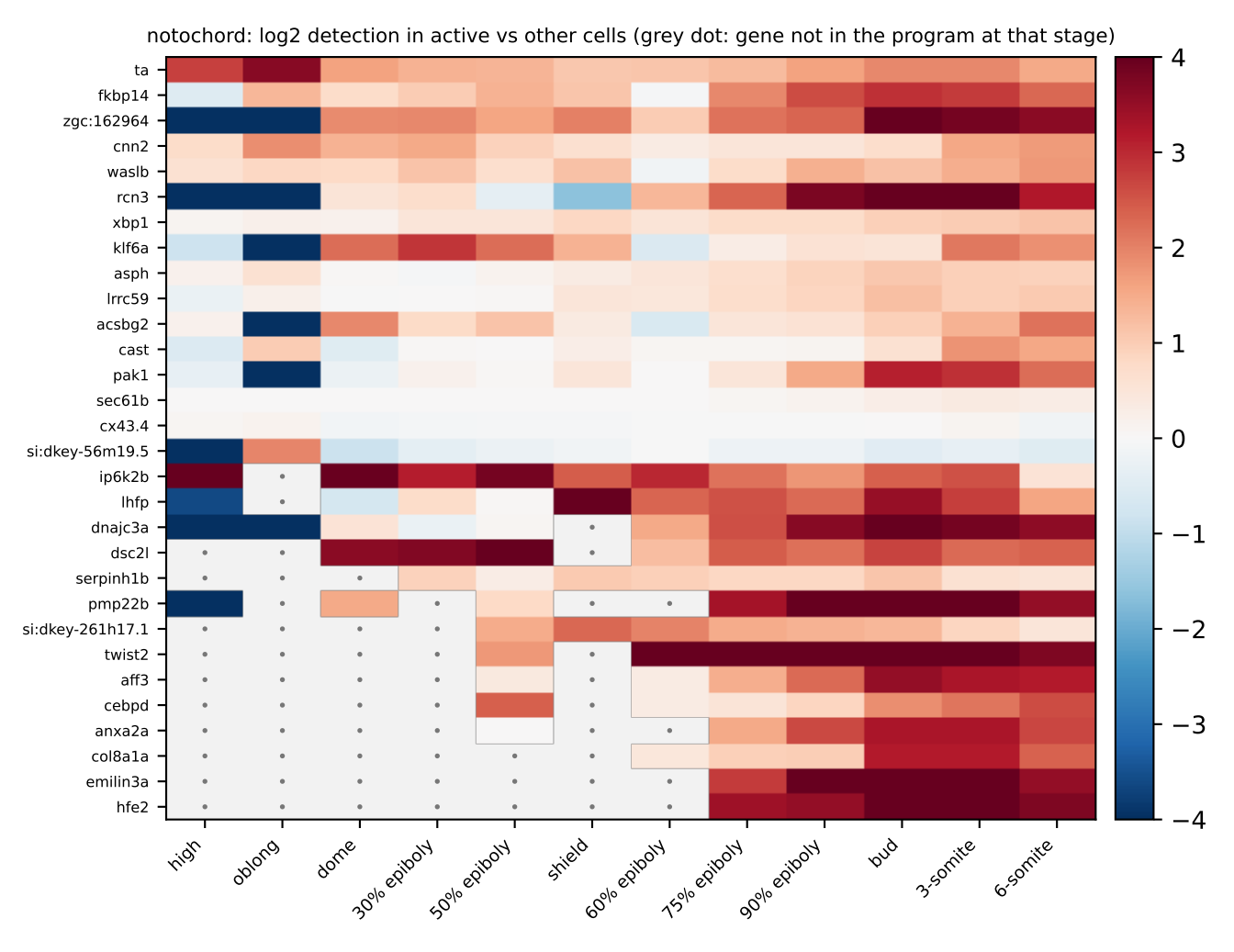

PGC: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
slc16a3
gra
ddx4
org
tdrd7a
dnd1
nanos3
ca15b
rgs14a
nanos1
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
2
0
−2
−4

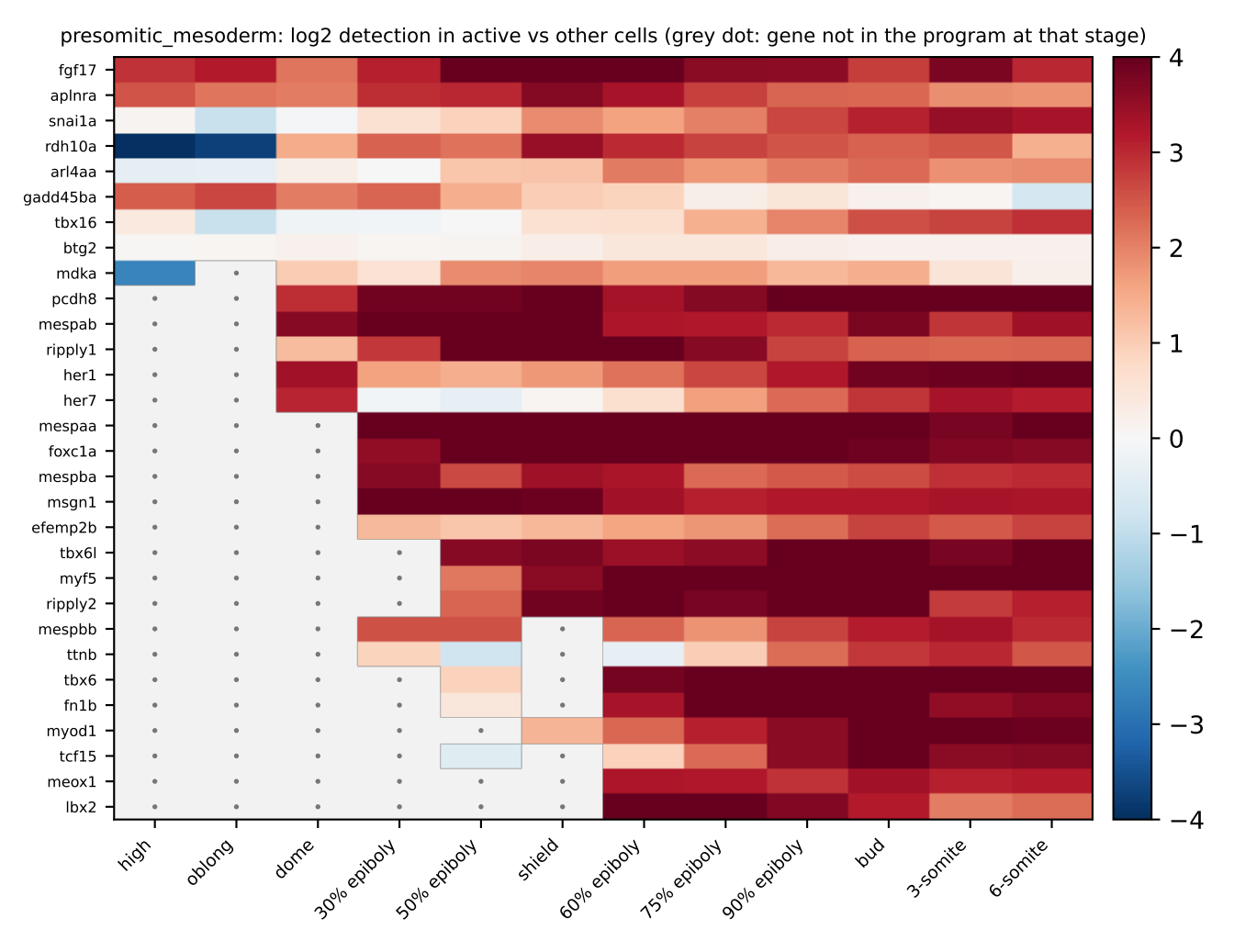

dorsal_organizer: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
gsc
noto
dact2
net1
acvr1ba
xbp1
fstl1b
rab32a
cbsb
mycb
smad7
insb
pitx2
kirrel3l
plod1a
twist2
creb3l1
plod2
cthrc1a
dharma
capn2b
igf2b
tbr1b
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
3
2
1
0
−1
−2
−3
−4

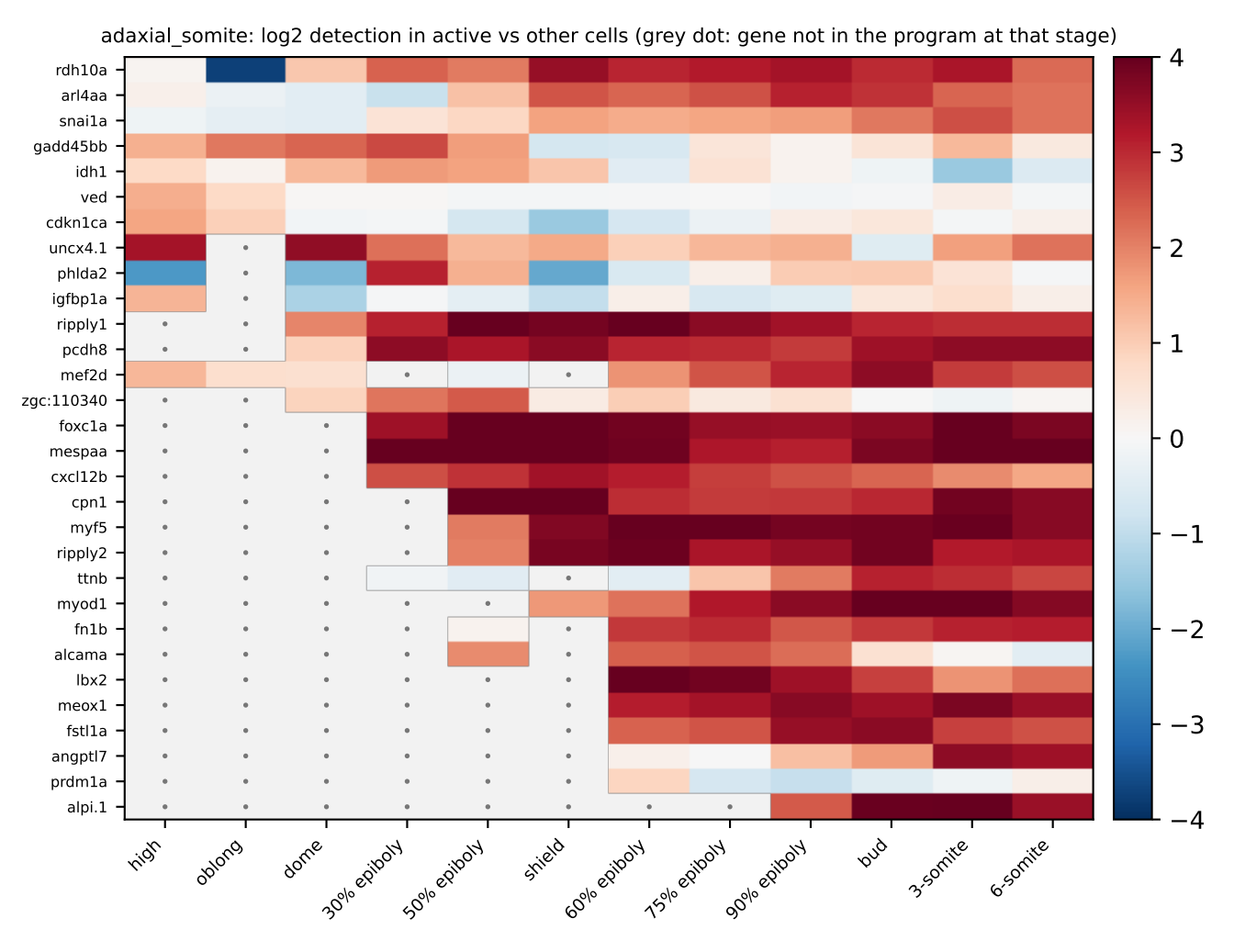

dorsal_forerunner: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
rassf7b
tjp3
pard6gb
rock2b
sult6b1
jupa
tmem106ba
vgll4l
mns1
prickle1a
dnmt3bb.1
zgc:55461
rasgef1ba
cxcr4a
si:ch211-152c2.3
pfkfb4b
apela
arg2
cfap45
cfap53
ca9
nnr
cdh2
si:ch73-234b20.5
relt
cldn5a
foxj1a
ribc2
tagln3b
ccdc151
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
3
2
1
0
−1
−2
−3
−4

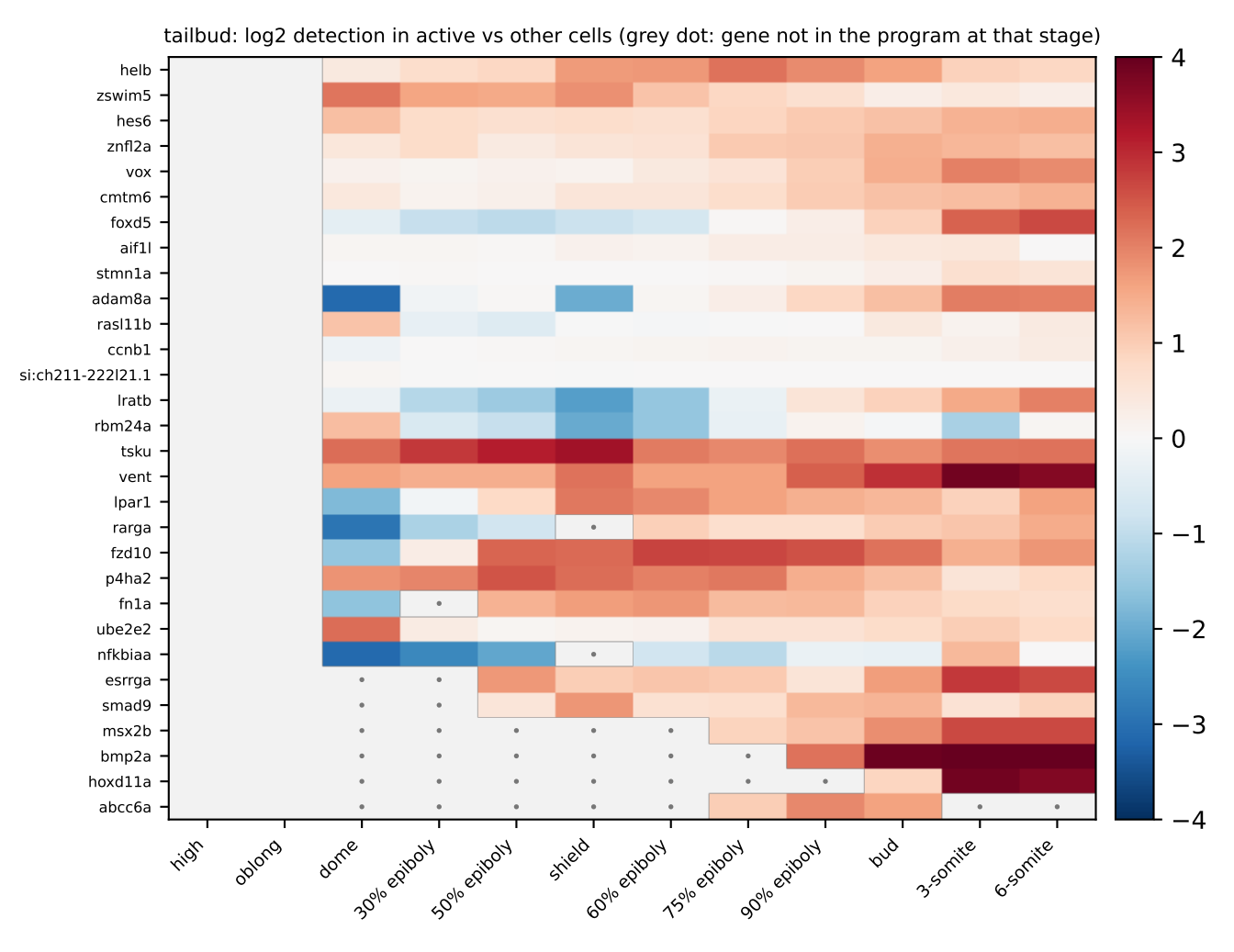

endoderm: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
sept9a
ackr3b
etv5a
gid8b
flrt3
nme4
foxa2
gata5
foxa3
mixl1
nipsnap1
nkx6.1
sox32
sox17
met
sebox
foxa1
fgfrl1b
nkx6.2
mnx1
mnx2b
col2a1a
bmp6
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
3
2
1
0
−1
−2
−3
−4

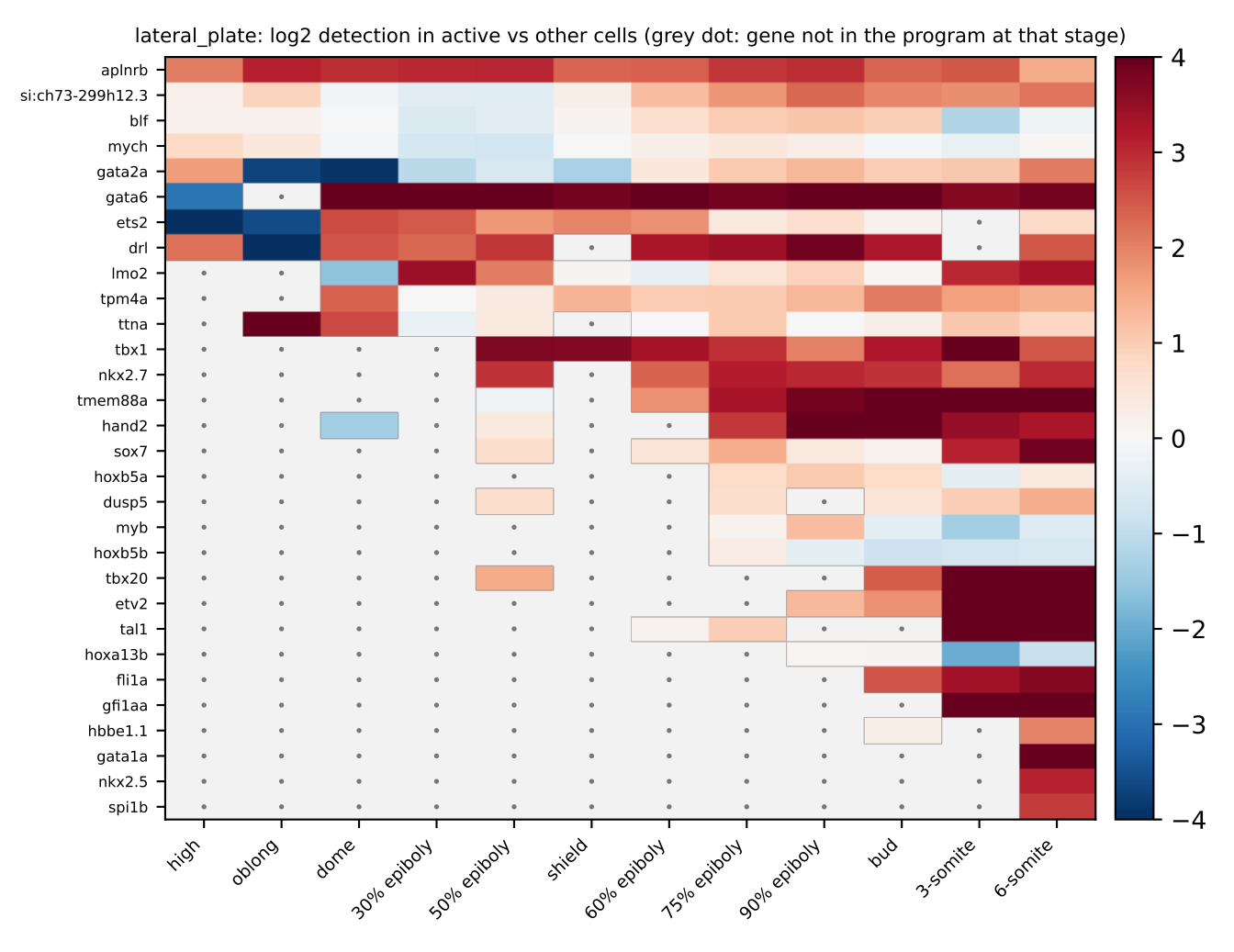

prechordal_plate: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
gsc
b3gnt5b
fam84a
fzd8a
ezrb
cd63
cdh1
hyou1
nav2b
asb11
gadd45ga
ddit4
si:dkey-27i16.2
mllt1b
tagln2
pitx2
elovl6
atp1b3a
icn
rdh8a
lmo1
tmem120a
nr2f6b
pcdh10b
si:ch211-163l21.8
ctslb
nkx2.2b
capn2b
nmt1b
inhbaa
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
3
2
1
0
−1
−2
−3
−4

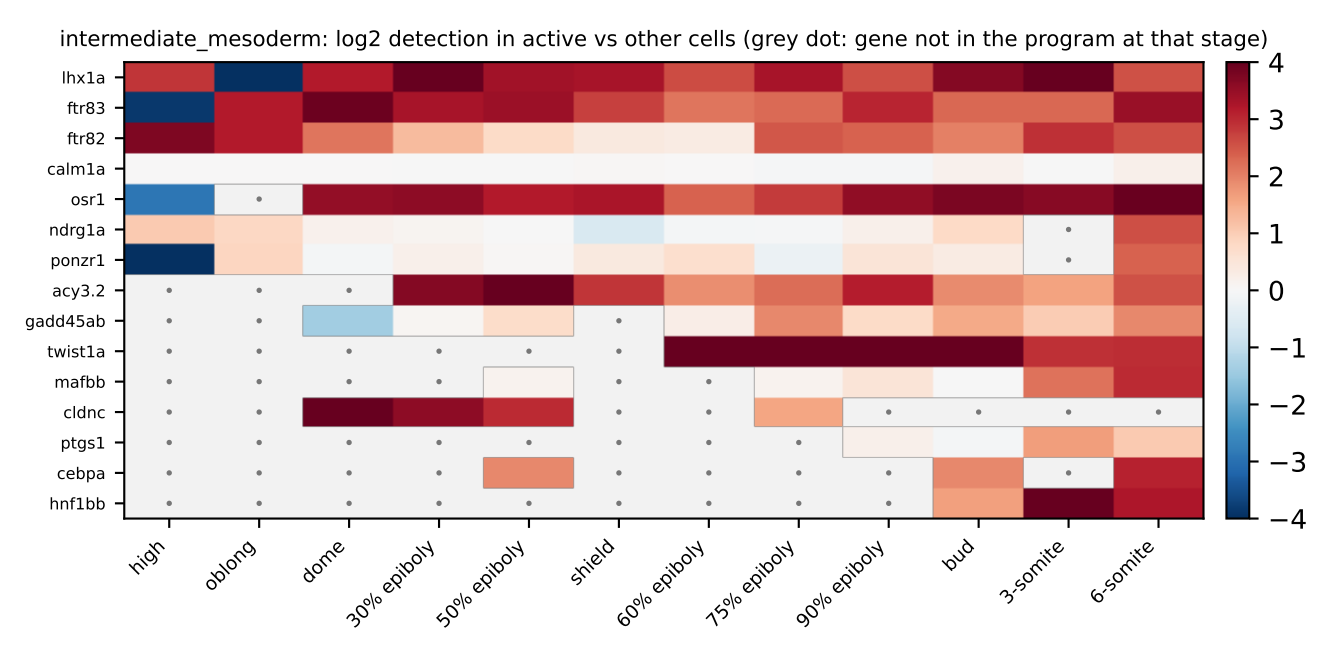

notochord: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
ta
fkbp14
zgc:162964
cnn2
waslb
rcn3
xbp1
klf6a
asph
lrrc59
acsbg2
cast
pak1
sec61b
cx43.4
si:dkey-56m19.5
ip6k2b
lhfp
dnajc3a
dsc2l
serpinh1b
pmp22b
si:dkey-261h17.1
twist2
aff3
cebpd
anxa2a
col8a1a
emilin3a
hfe2
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
3
2
1
0
−1
−2
−3
−4

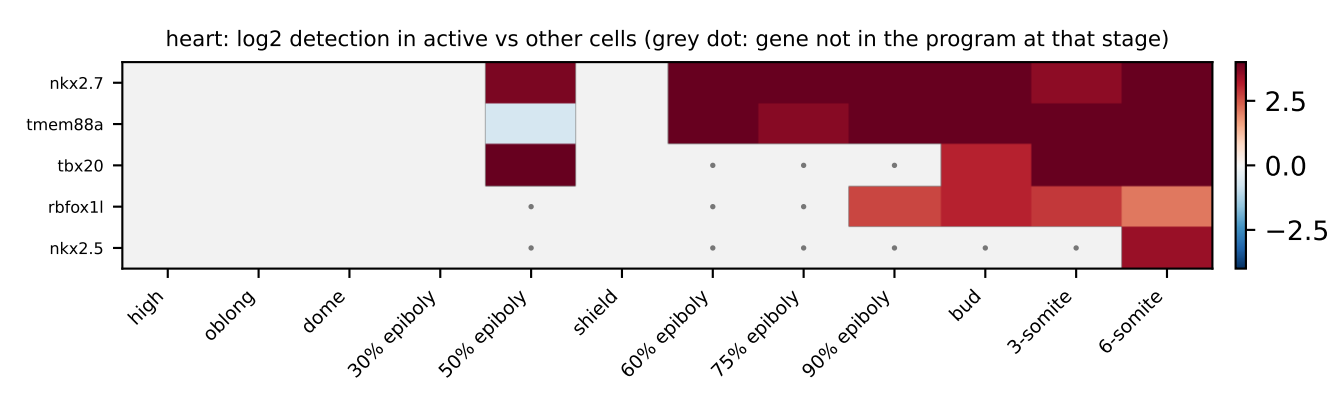

presomitic_mesoderm: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
fgf17
aplnra
snai1a
rdh10a
arl4aa
gadd45ba
tbx16
btg2
mdka
pcdh8
mespab
ripply1
her1
her7
mespaa
foxc1a
mespba
msgn1
efemp2b
tbx6l
myf5
ripply2
mespbb
ttnb
tbx6
fn1b
myod1
tcf15
meox1
lbx2
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
3
2
1
0
−1
−2
−3
−4

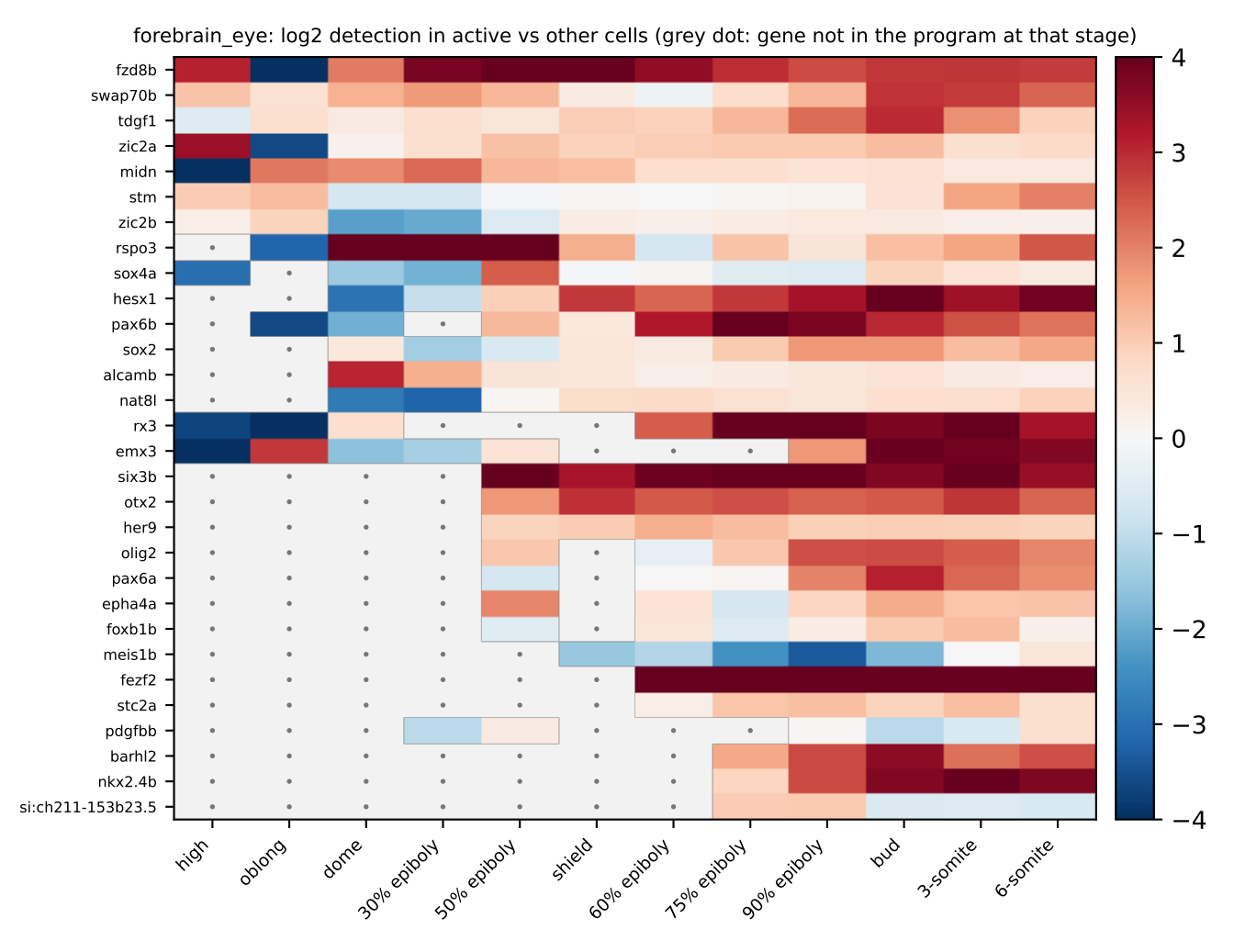

adaxial_somite: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
rdh10a
arl4aa
snai1a
gadd45bb
idh1
ved
cdkn1ca
uncx4.1
phlda2
igfbp1a
ripply1
pcdh8
mef2d
zgc:110340
foxc1a
mespaa
cxcl12b
cpn1
myf5
ripply2
ttnb
myod1
fn1b
alcama
lbx2
meox1
fstl1a
angptl7
prdm1a
alpi.1
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
3
2
1
0
−1
−2
−3
−4

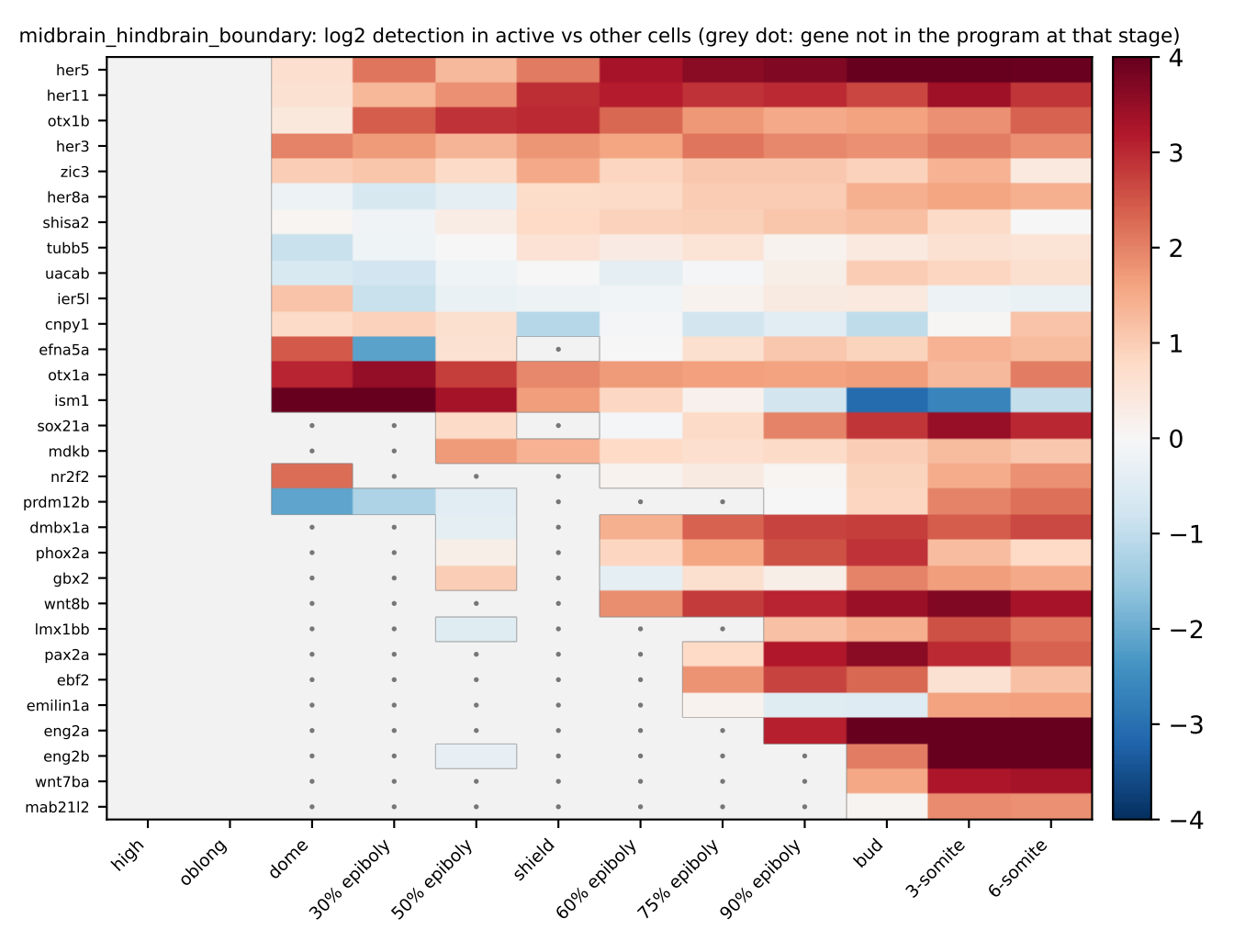

tailbud: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
helb
zswim5
hes6
znfl2a
vox
cmtm6
foxd5
aif1l
stmn1a
adam8a
rasl11b
ccnb1
si:ch211-222l21.1
lratb
rbm24a
tsku
vent
lpar1
rarga
fzd10
p4ha2
fn1a
ube2e2
nfkbiaa
esrrga
smad9
msx2b
bmp2a
hoxd11a
abcc6a
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
3
2
1
0
−1
−2
−3
−4

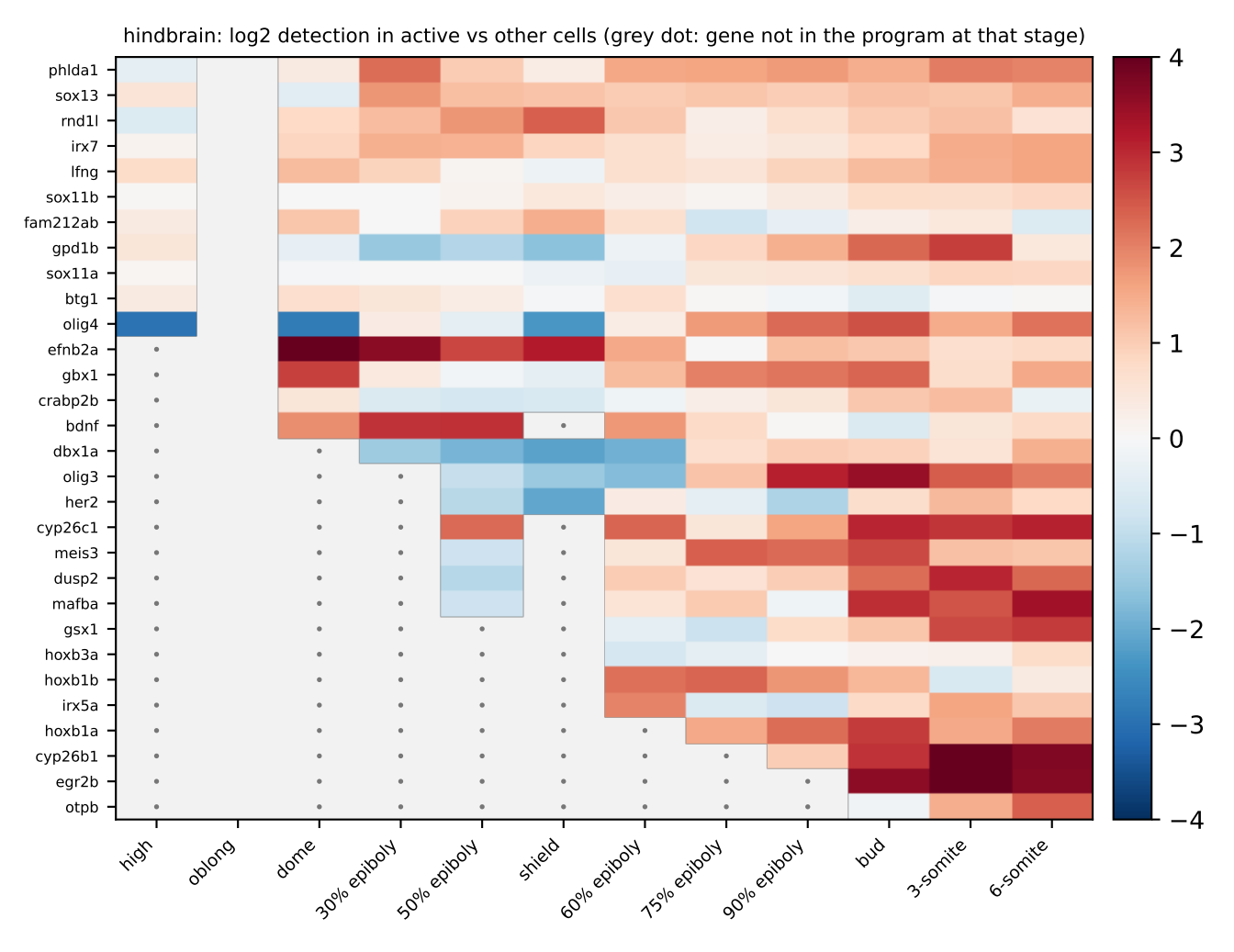

lateral_plate: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
aplnrb
si:ch73-299h12.3
blf
mych
gata2a
gata6
ets2
drl
lmo2
tpm4a
ttna
tbx1
nkx2.7
tmem88a
hand2
sox7
hoxb5a
dusp5
myb
hoxb5b
tbx20
etv2
tal1
hoxa13b
fli1a
gfi1aa
hbbe1.1
gata1a
nkx2.5
spi1b
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
3
2
1
0
−1
−2
−3
−4

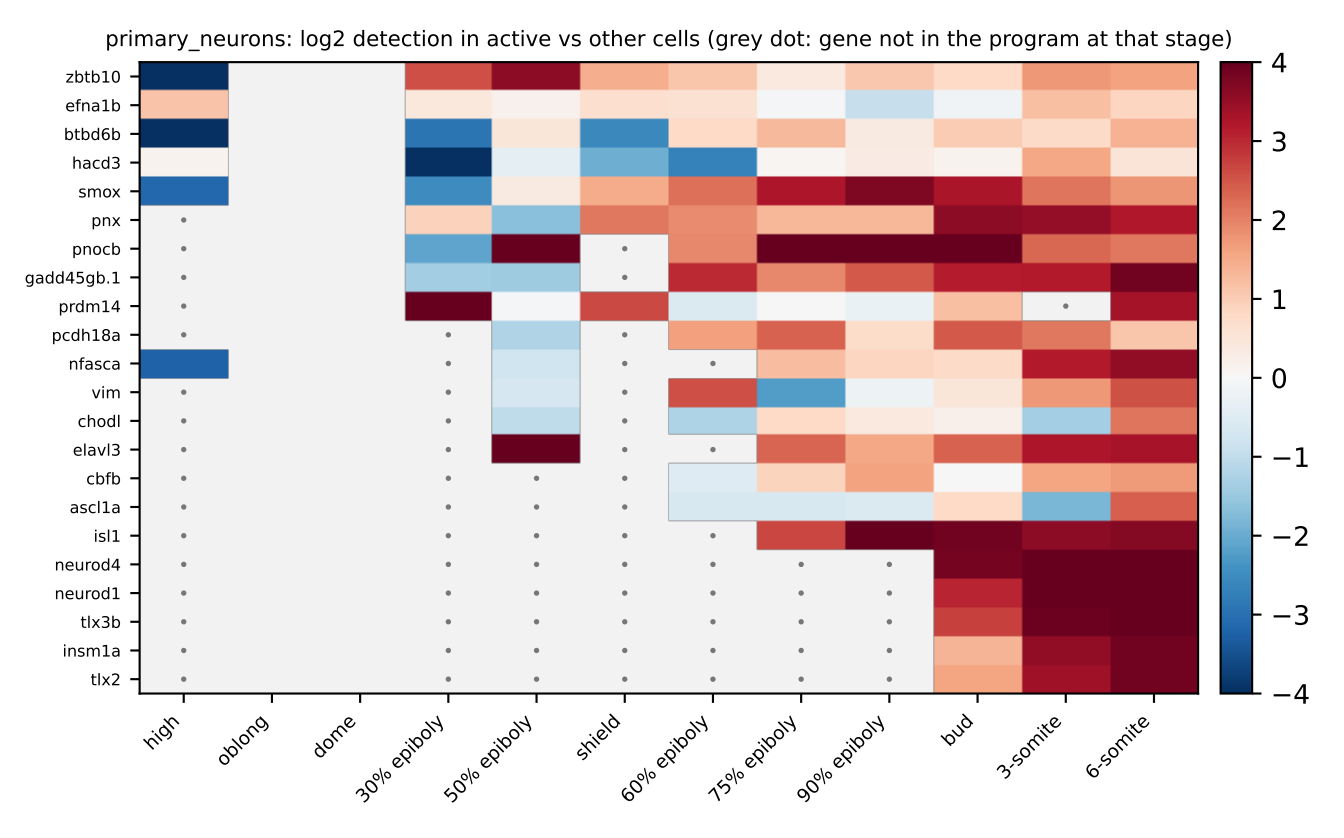

intermediate_mesoderm: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
lhx1a
ftr83
ftr82
calm1a
osr1
ndrg1a
ponzr1
acy3.2
gadd45ab
twist1a
mafbb
cldnc
ptgs1
cebpa
hnf1bb
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
3
2
1
0
−1
−2
−3
−4

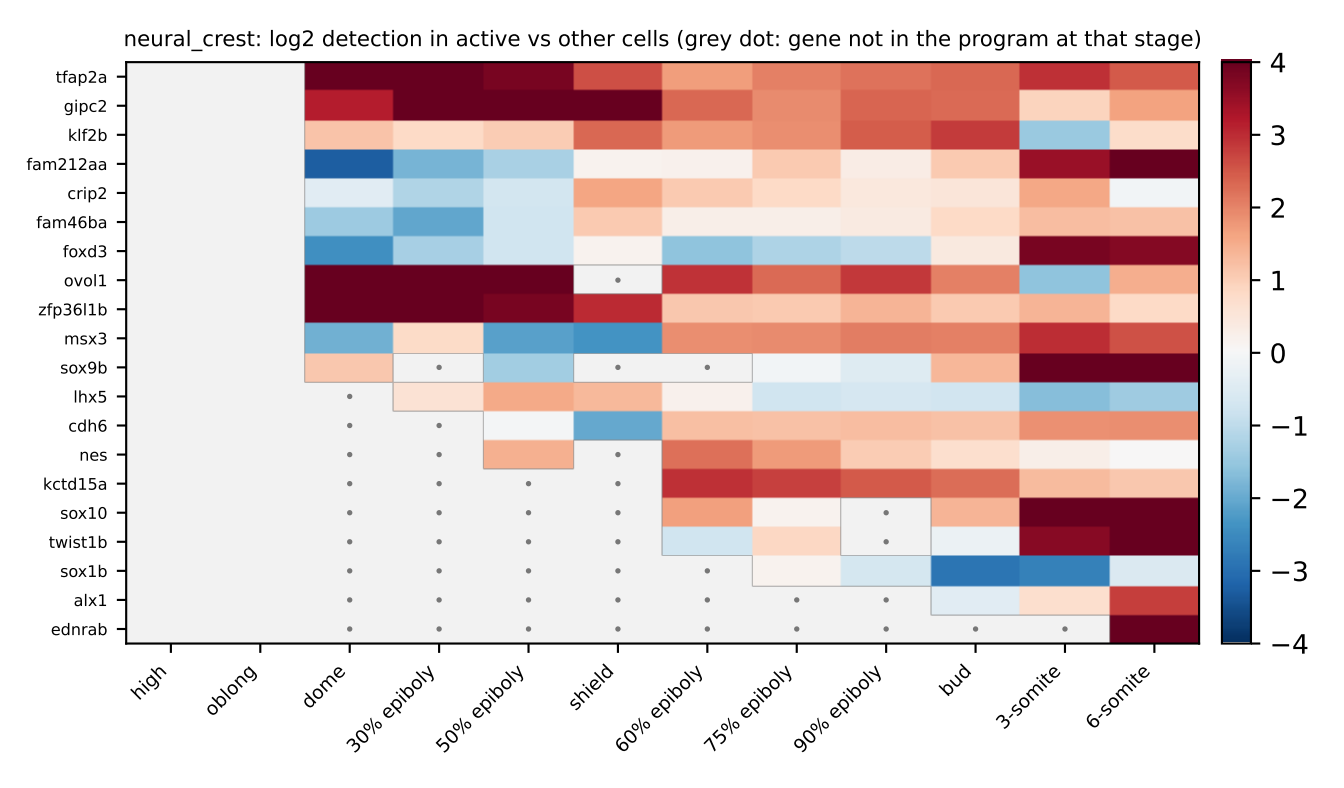

heart: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
nkx2.7
tmem88a
tbx20
rbfox1l
nkx2.5
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
2.5
0.0
−2.5

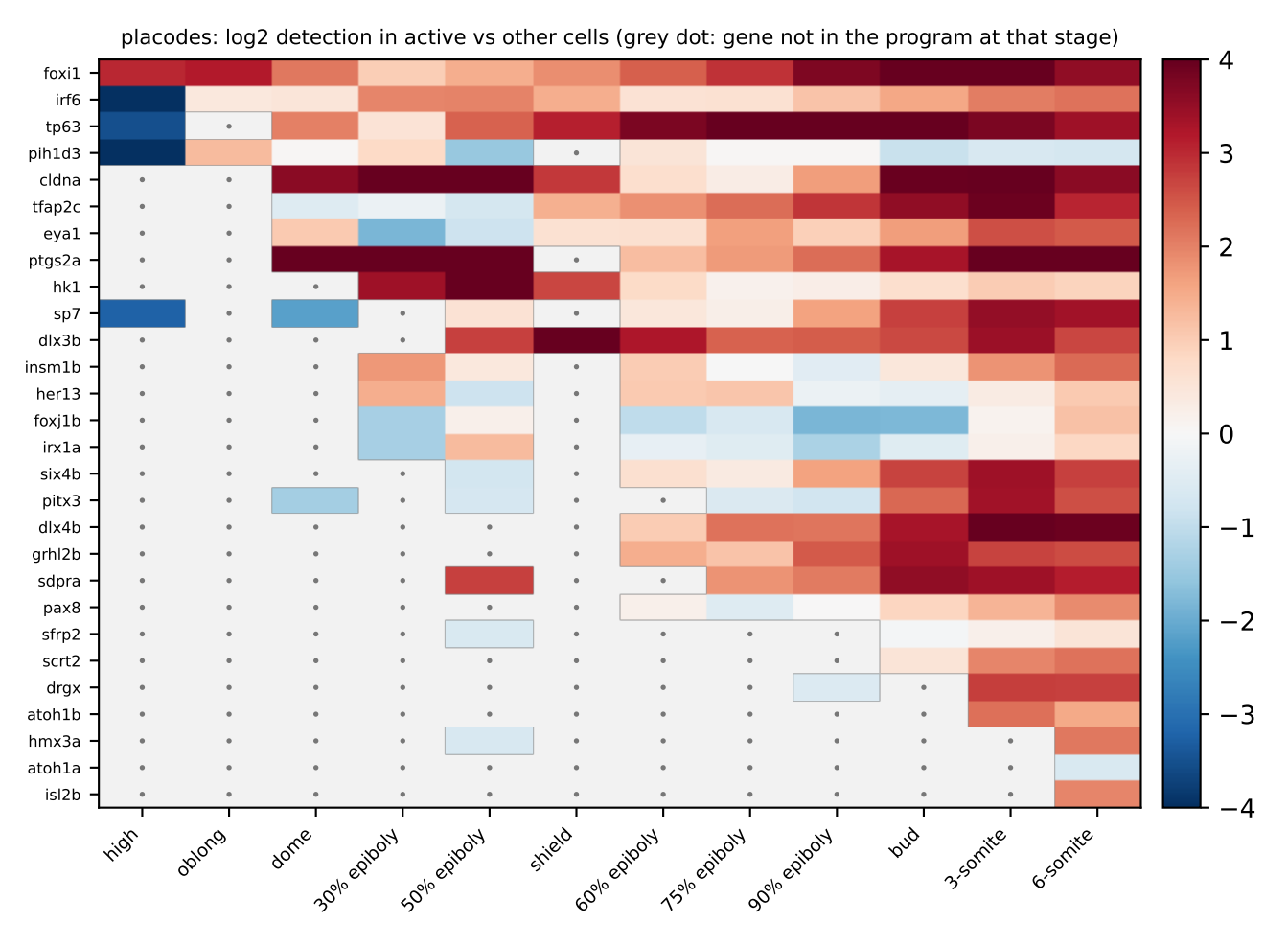

forebrain_eye: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
fzd8b
swap70b
tdgf1
zic2a
midn
stm
zic2b
rspo3
sox4a
hesx1
pax6b
sox2
alcamb
nat8l
rx3
emx3
six3b
otx2
her9
olig2
pax6a
epha4a
foxb1b
meis1b
fezf2
stc2a
pdgfbb
barhl2
nkx2.4b
si:ch211-153b23.5
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
3
2
1
0
−1
−2
−3
−4

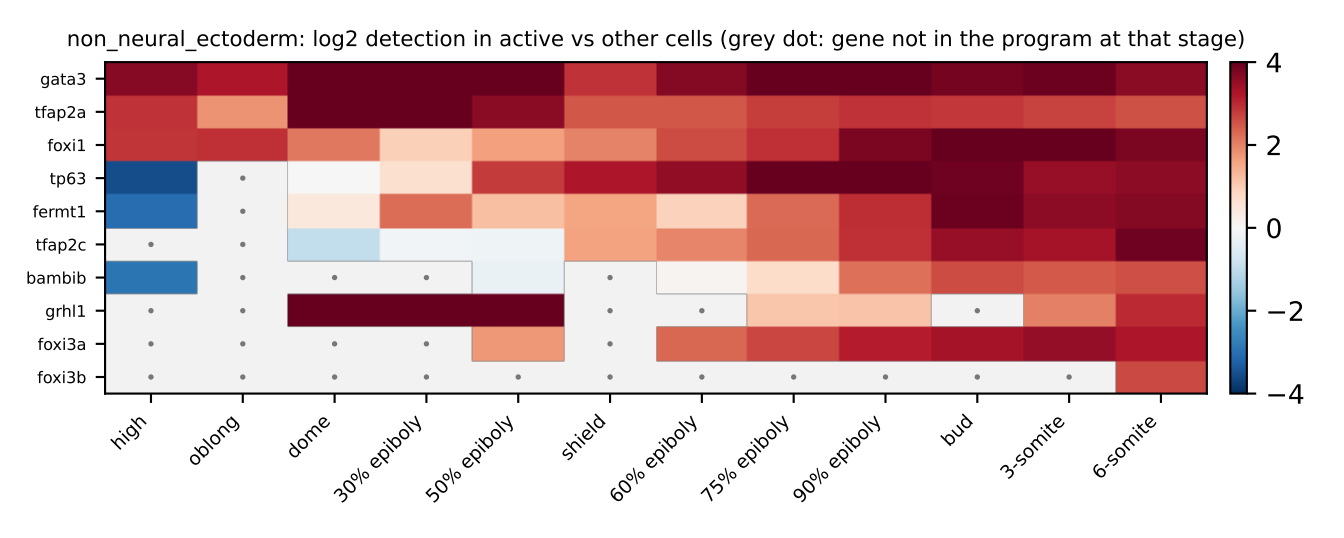

midbrain_hindbrain_boundary: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
her5
her11
otx1b
her3
zic3
her8a
shisa2
tubb5
uacab
ier5l
cnpy1
efna5a
otx1a
ism1
sox21a
mdkb
nr2f2
prdm12b
dmbx1a
phox2a
gbx2
wnt8b
lmx1bb
pax2a
ebf2
emilin1a
eng2a
eng2b
wnt7ba
mab21l2
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
3
2
1
0
−1
−2
−3
−4

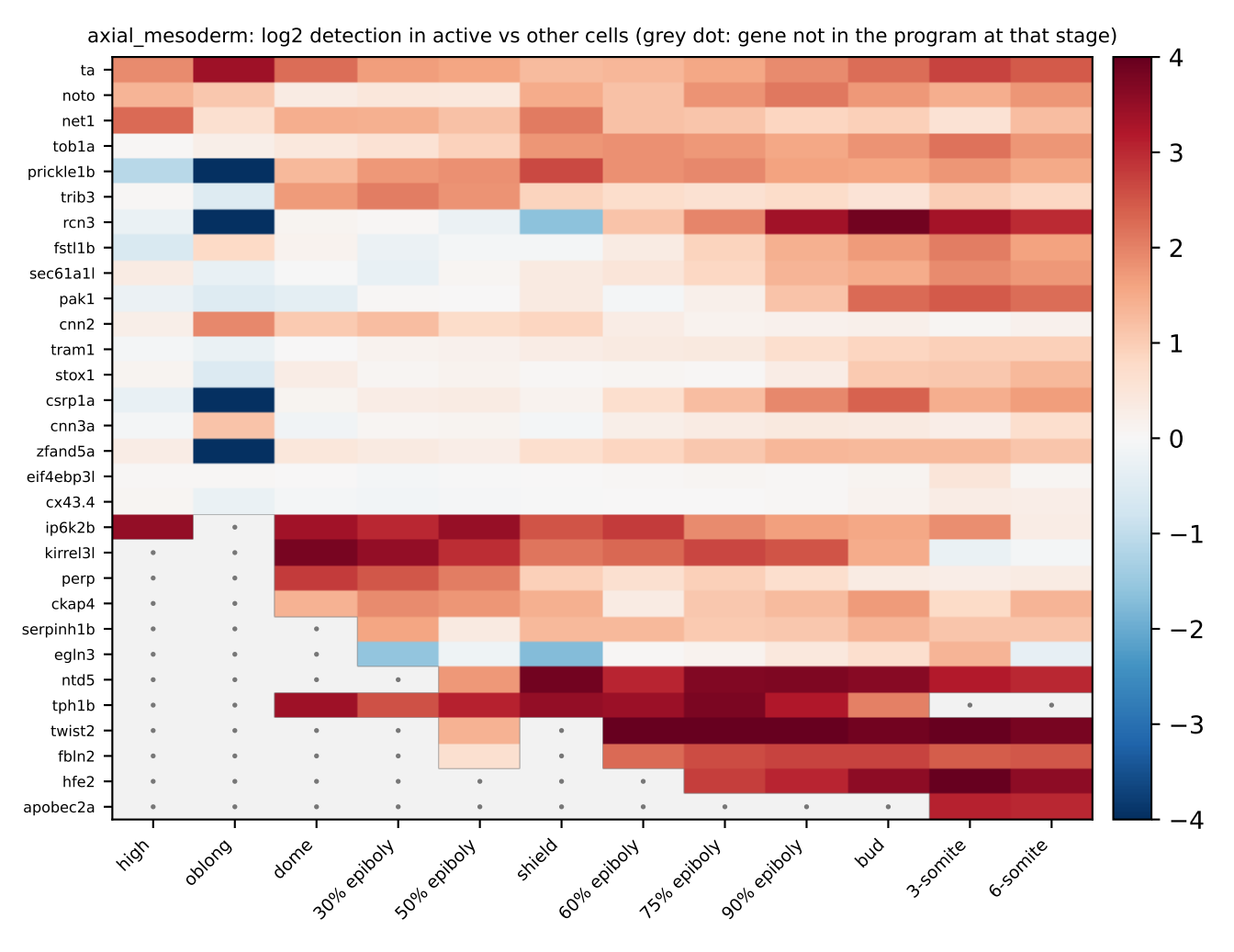

hindbrain: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
phlda1
sox13
rnd1l
irx7
lfng
sox11b
fam212ab
gpd1b
sox11a
btg1
olig4
efnb2a
gbx1
crabp2b
bdnf
dbx1a
olig3
her2
cyp26c1
meis3
dusp2
mafba
gsx1
hoxb3a
hoxb1b
irx5a
hoxb1a
cyp26b1
egr2b
otpb
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
3
2
1
0
−1
−2
−3
−4

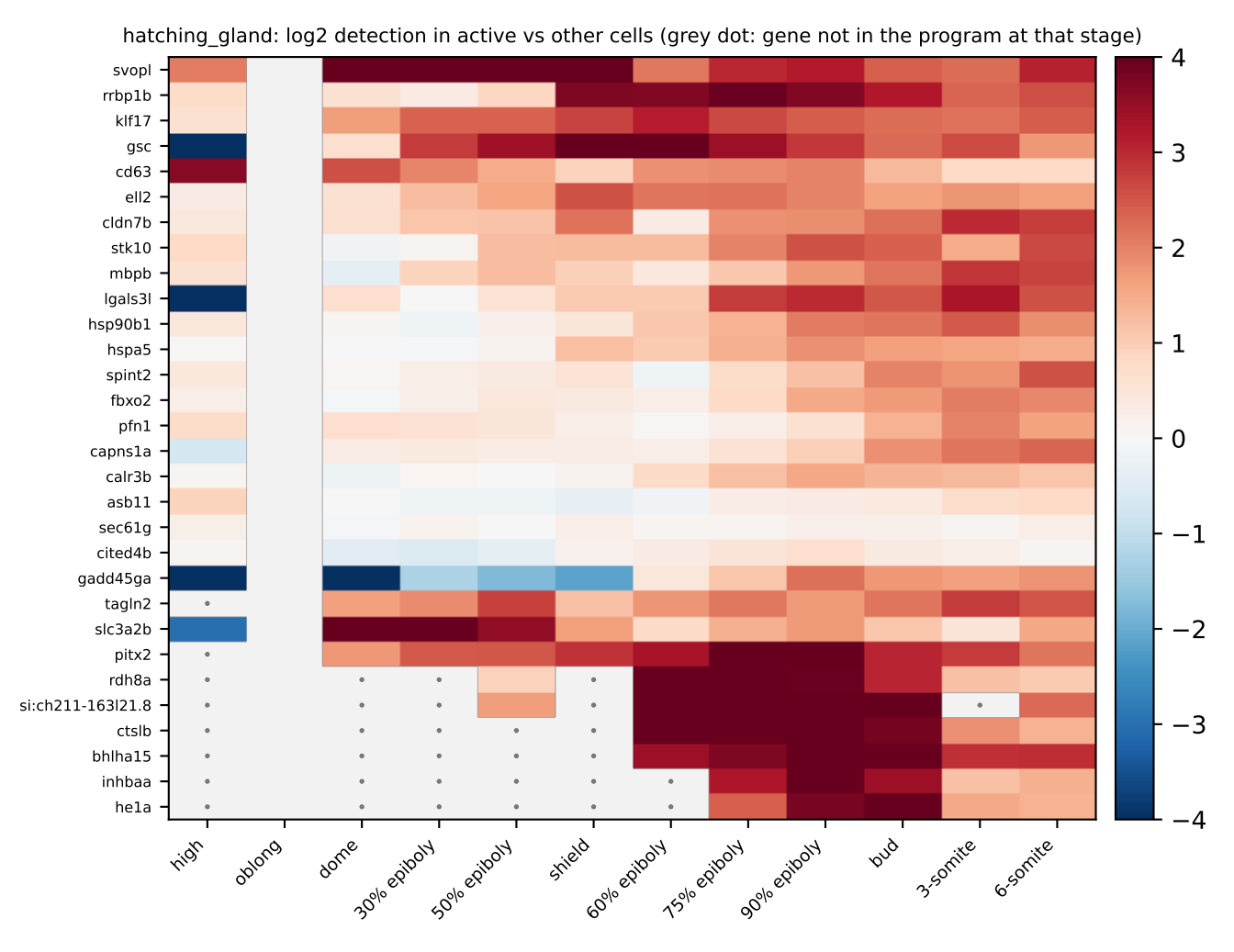

primary_neurons: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
zbtb10
efna1b
btbd6b
hacd3
smox
pnx
pnocb
gadd45gb.1
prdm14
pcdh18a
nfasca
vim
chodl
elavl3
cbfb
ascl1a
isl1
neurod4
neurod1
tlx3b
insm1a
tlx2
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
3
2
1
0
−1
−2
−3
−4

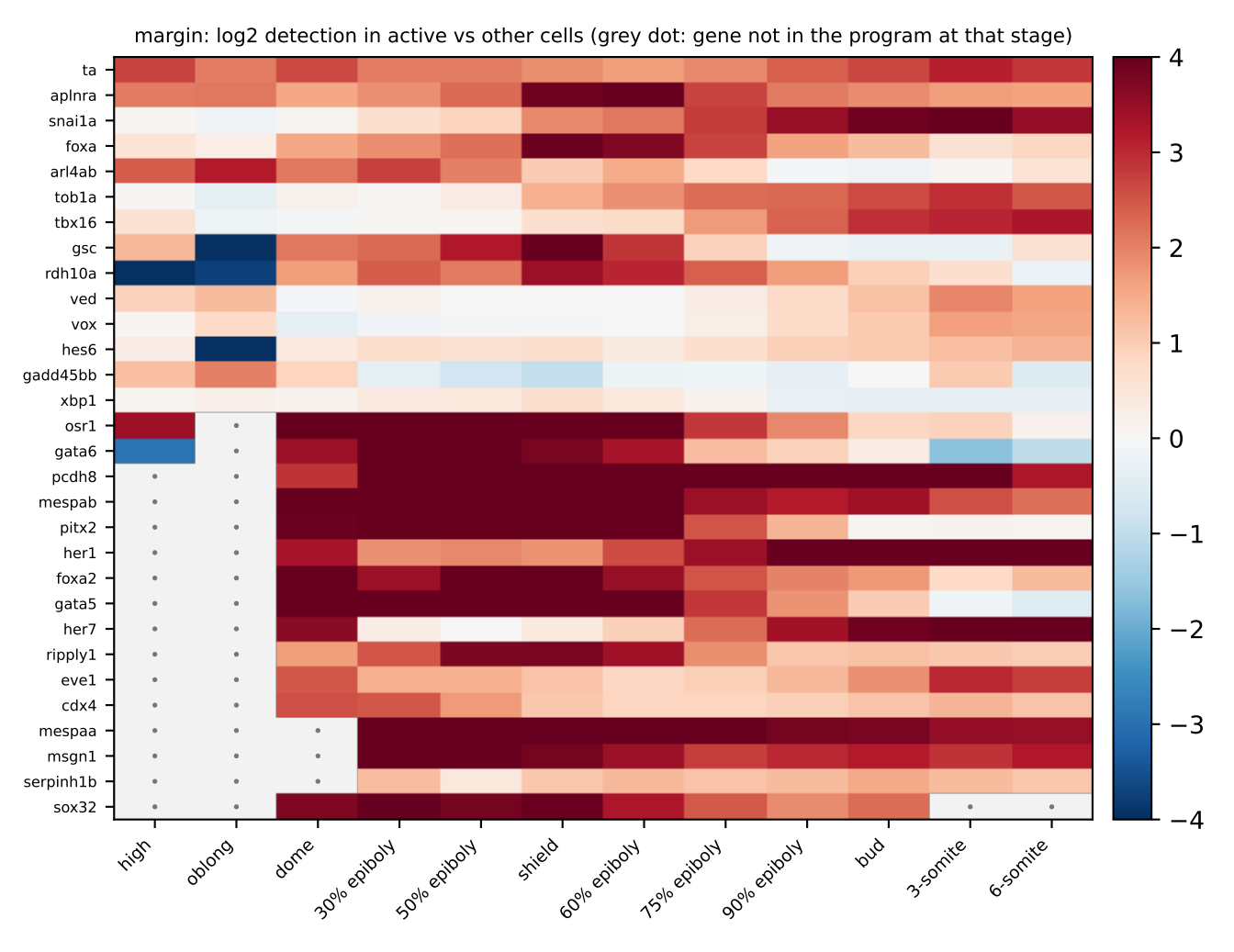

neural_crest: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
tfap2a
gipc2
klf2b
fam212aa
crip2
fam46ba
foxd3
ovol1
zfp36l1b
msx3
sox9b
lhx5
cdh6
nes
kctd15a
sox10
twist1b
sox1b
alx1
ednrab
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
3
2
1
0
−1
−2
−3
−4

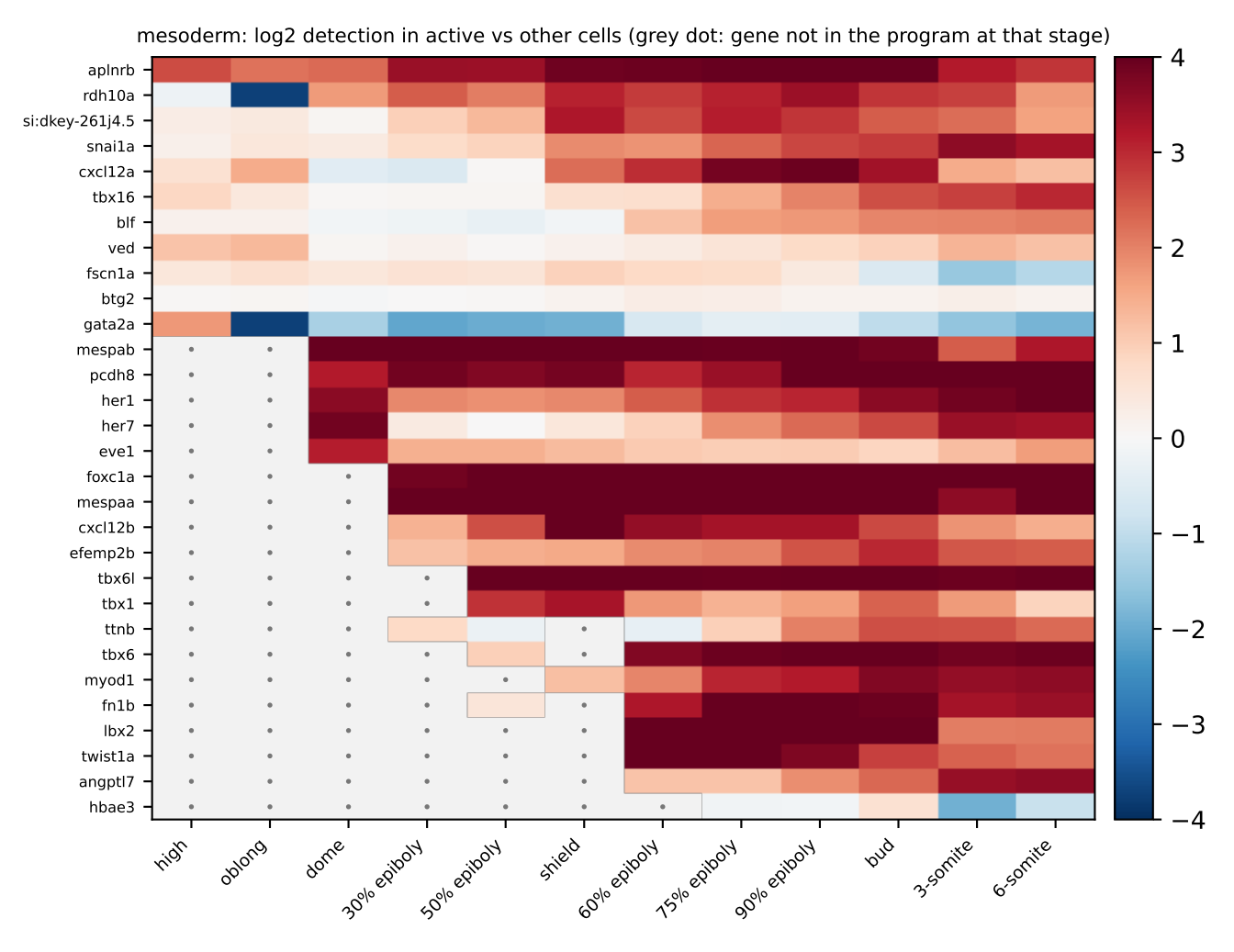

placodes: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
foxi1
irf6
tp63
pih1d3
cldna
tfap2c
eya1
ptgs2a
hk1
sp7
dlx3b
insm1b
her13
foxj1b
irx1a
six4b
pitx3
dlx4b
grhl2b
sdpra
pax8
sfrp2
scrt2
drgx
atoh1b
hmx3a
atoh1a
isl2b
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
3
2
1
0
−1
−2
−3
−4

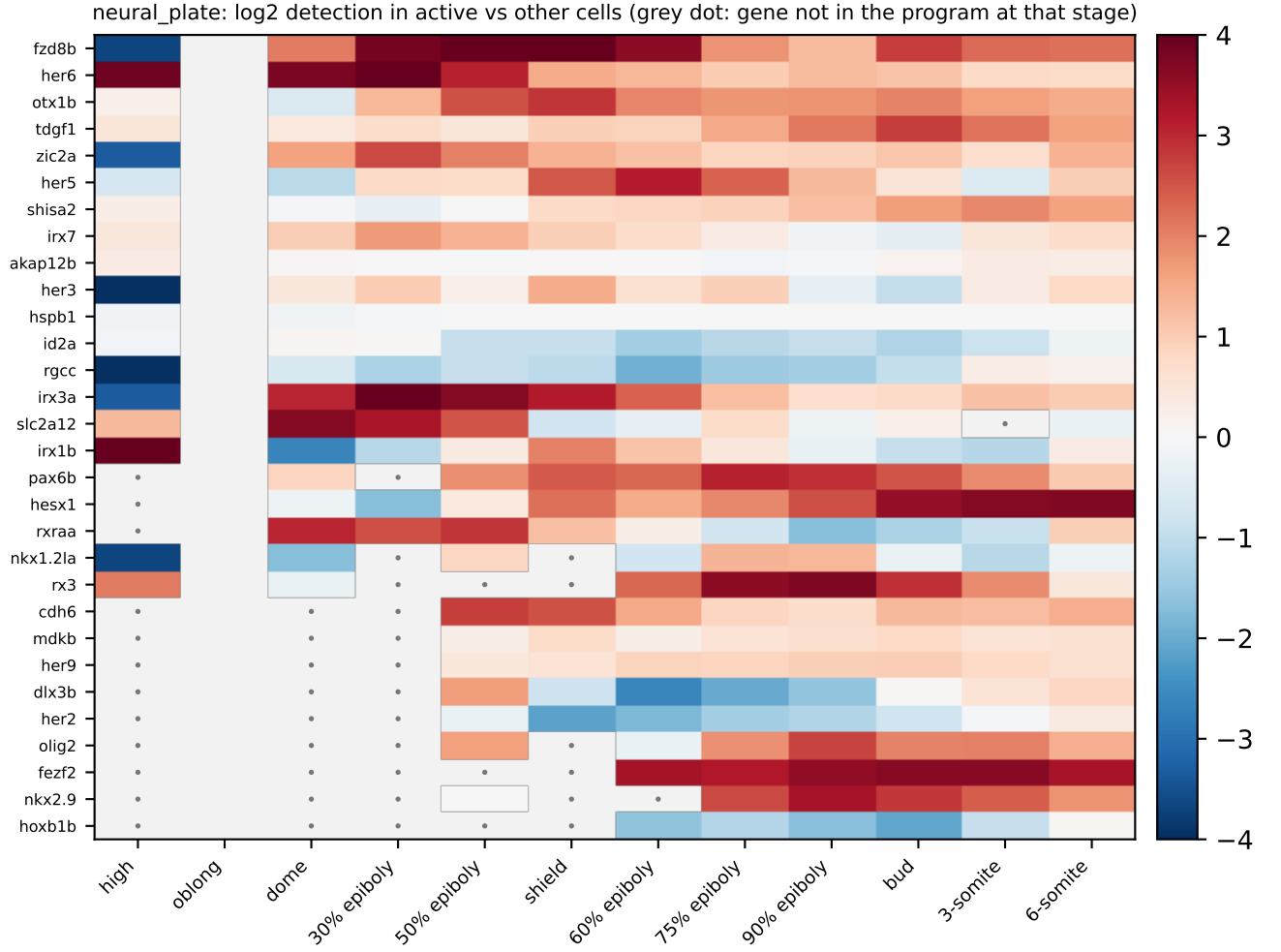

non_neural_ectoderm: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
gata3
tfap2a
foxi1
tp63
fermt1
tfap2c
bambib
grhl1
foxi3a
foxi3b
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
2
0
−2
−4

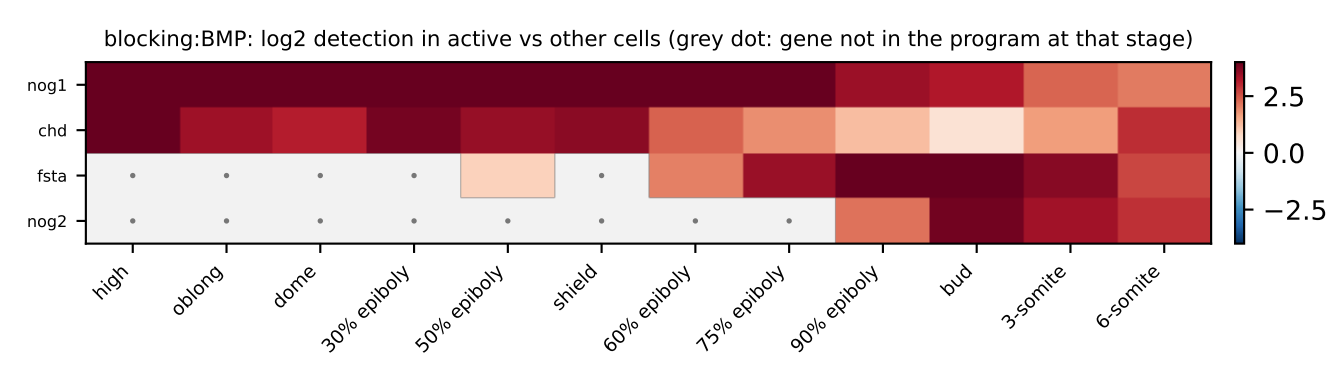

axial_mesoderm: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
ta
noto
net1
tob1a
prickle1b
trib3
rcn3
fstl1b
sec61a1l
pak1
cnn2
tram1
stox1
csrp1a
cnn3a
zfand5a
eif4ebp3l
cx43.4
ip6k2b
kirrel3l
perp
ckap4
serpinh1b
egln3
ntd5
tph1b
twist2
fbln2
hfe2
apobec2a
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
3
2
1
0
−1
−2
−3
−4

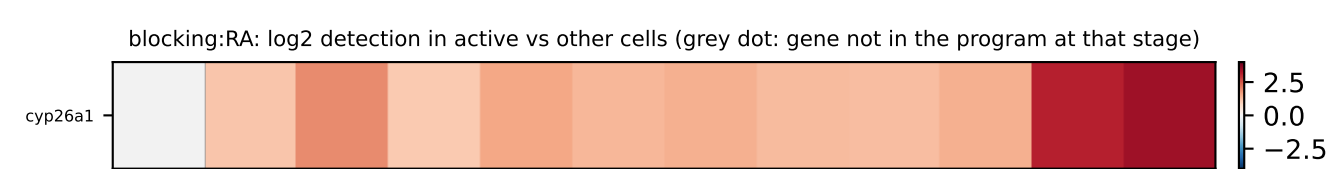

hatching_gland: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
svopl
rrbp1b
klf17
gsc
cd63
ell2
cldn7b
stk10
mbpb
lgals3l
hsp90b1
hspa5
spint2
fbxo2
pfn1
capns1a
calr3b
asb11
sec61g
cited4b
gadd45ga
tagln2
slc3a2b
pitx2
rdh8a
si:ch211-163l21.8
ctslb
bhlha15
inhbaa
he1a
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
3
2
1
0
−1
−2
−3
−4

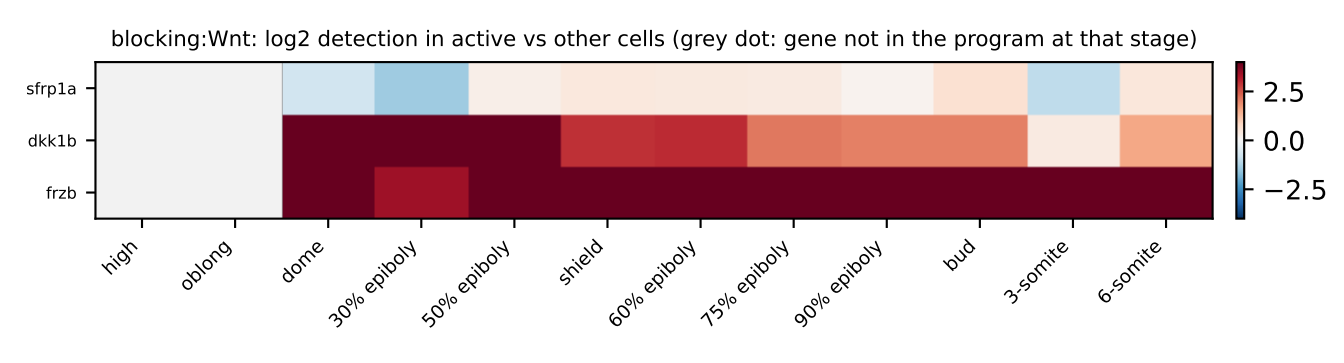

margin: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
ta
aplnra
snai1a
foxa
arl4ab
tob1a
tbx16
gsc
rdh10a
ved
vox
hes6
gadd45bb
xbp1
osr1
gata6
pcdh8
mespab
pitx2
her1
foxa2
gata5
her7
ripply1
eve1
cdx4
mespaa
msgn1
serpinh1b
sox32
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
3
2
1
0
−1
−2
−3
−4

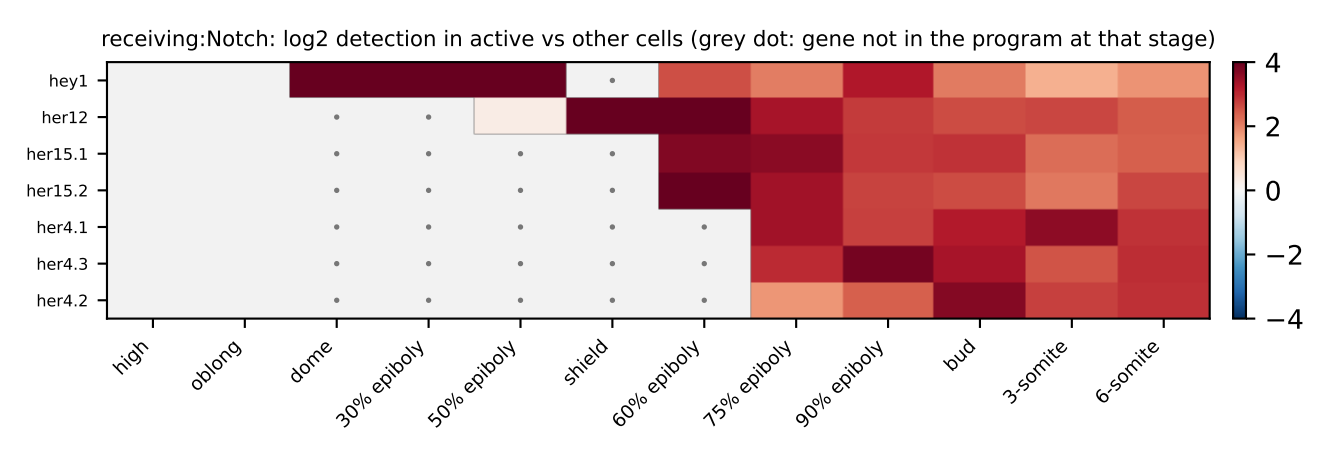

mesoderm: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
aplnrb
rdh10a
si:dkey-261j4.5
snai1a
cxcl12a
tbx16
blf
ved
fscn1a
btg2
gata2a
mespab
pcdh8
her1
her7
eve1
foxc1a
mespaa
cxcl12b
efemp2b
tbx6l
tbx1
ttnb
tbx6
myod1
fn1b
lbx2
twist1a
angptl7
hbae3
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
3
2
1
0
−1
−2
−3
−4

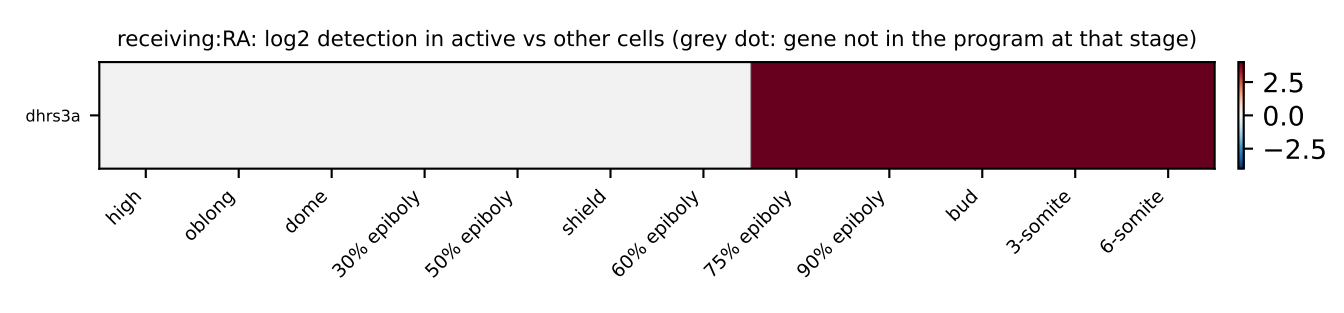

neural_plate: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
fzd8b
her6
otx1b
tdgf1
zic2a
her5
shisa2
irx7
akap12b
her3
hspb1
id2a
rgcc
irx3a
slc2a12
irx1b
pax6b
hesx1
rxraa
nkx1.2la
rx3
cdh6
mdkb
her9
dlx3b
her2
olig2
fezf2
nkx2.9
hoxb1b
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
3
2
1
0
−1
−2
−3
−4


blocking:BMP: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
nog1
chd
fsta
nog2
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
2.5
0.0
−2.5

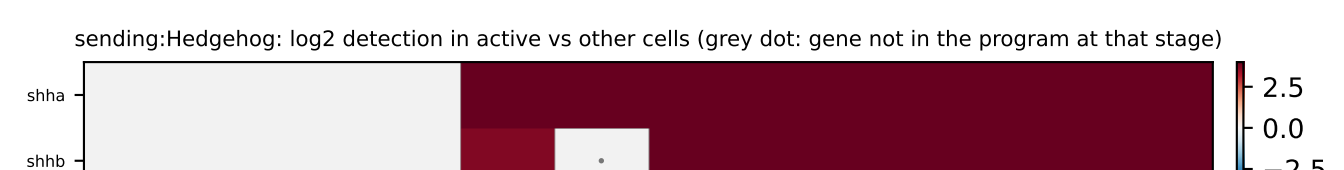

blocking:RA: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
cyp26a1
2.5
0.0
−2.5
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite

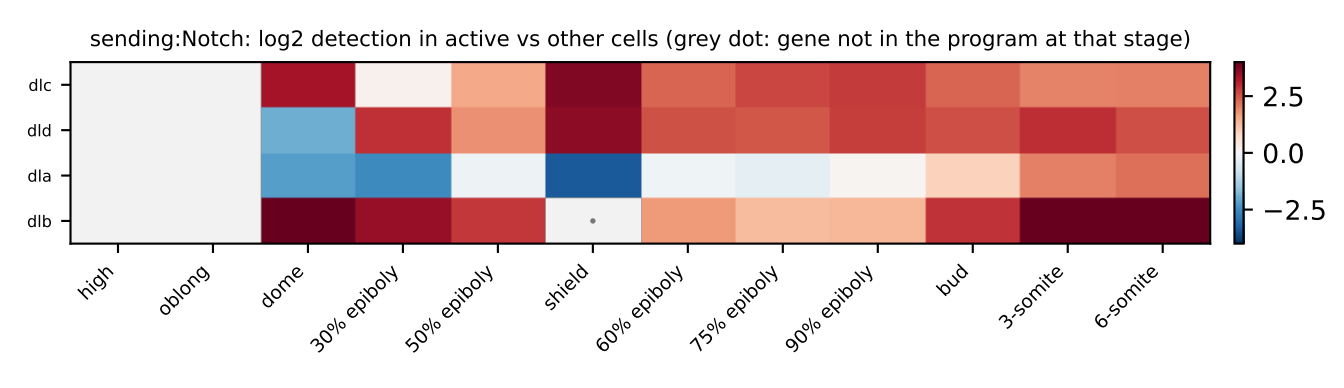

blocking:Wnt: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
sfrp1a
dkk1b
frzb
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
2.5
0.0
−2.5


receiving:Notch: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
hey1
her12
her15.1
her15.2
her4.1
her4.3
her4.2
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
2
0
−2
−4

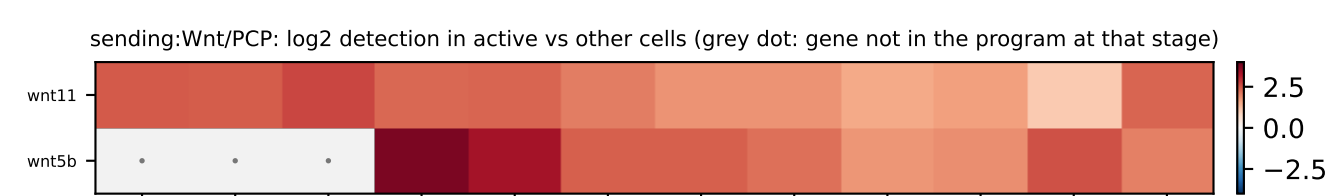

receiving:RA: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
dhrs3a
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
2.5
0.0
−2.5

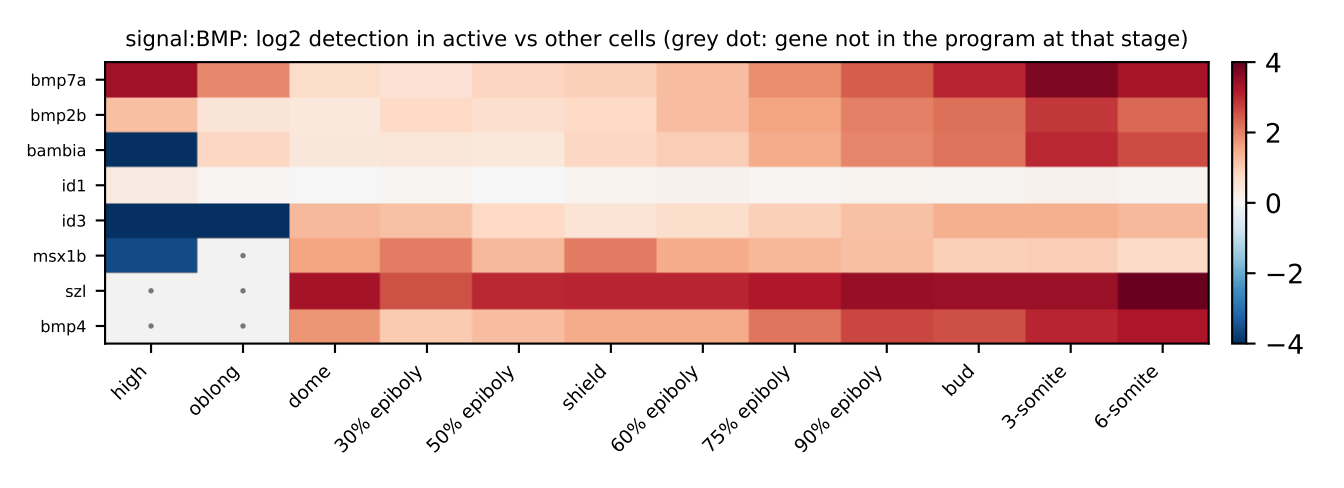

sending:BMP_organizer: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
admp
2.5
0.0
−2.5
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite

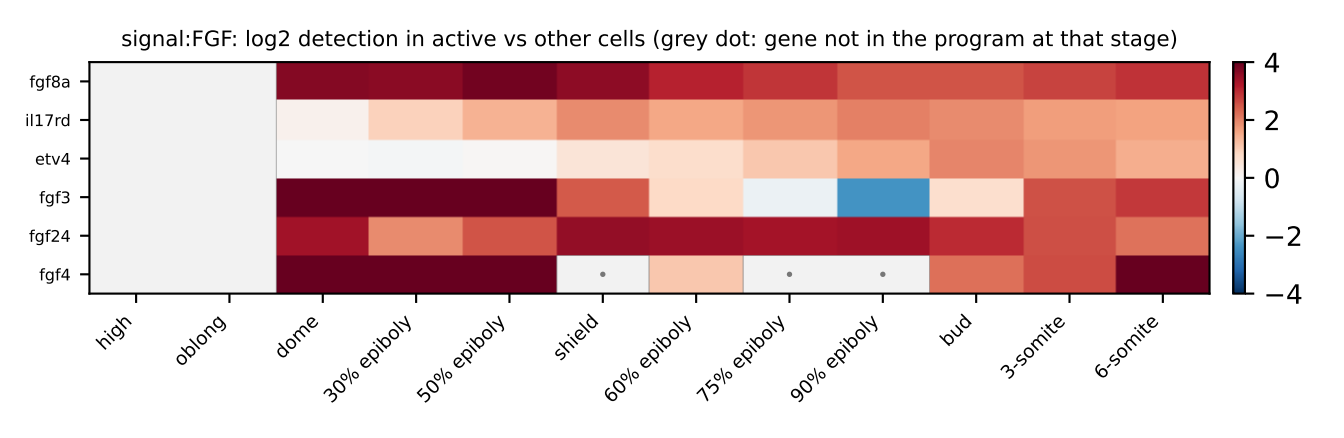

sending:Hedgehog: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
shha
shhb
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
2.5
0.0
−2.5

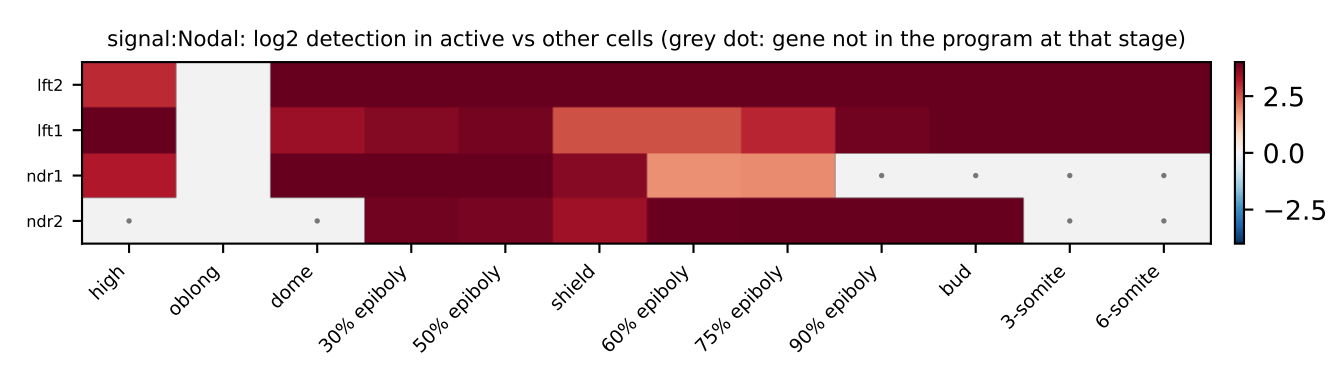

sending:Notch: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
dlc
dld
dla
dlb
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
2.5
0.0
−2.5

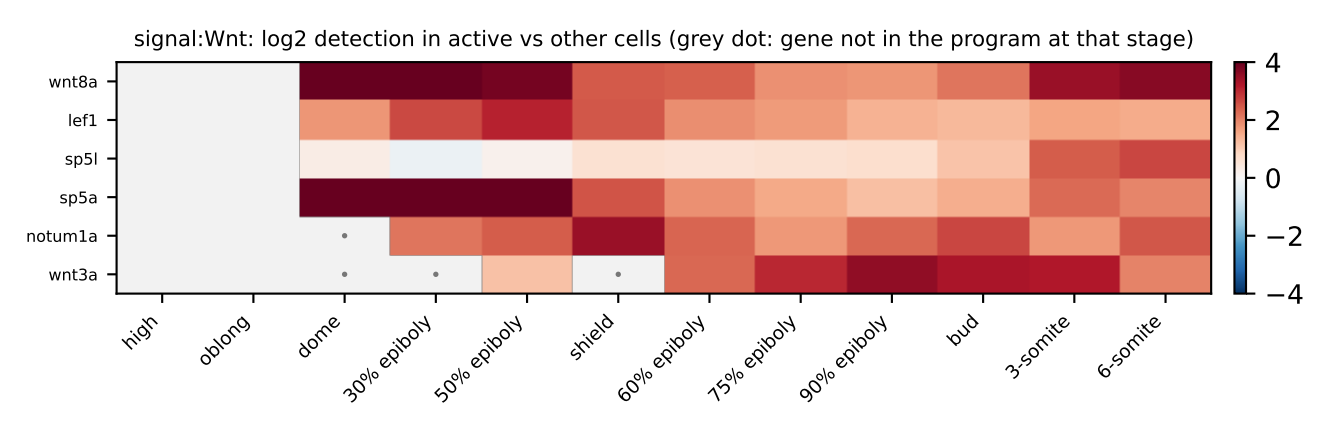

sending:RA: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
aldh1a2
2.5
0.0
−2.5
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite

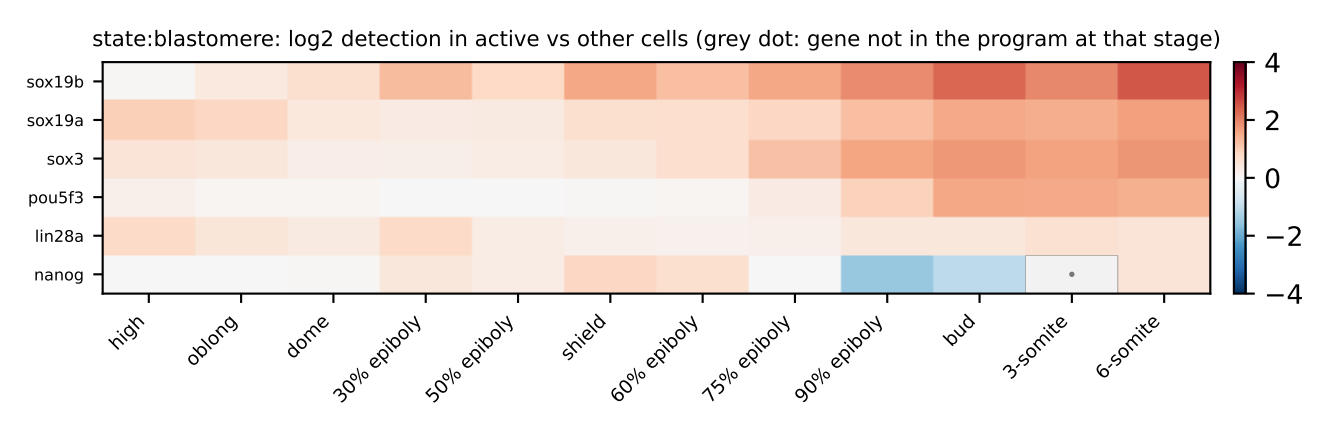

sending:Wnt/PCP: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
wnt11
wnt5b
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
2.5
0.0
−2.5

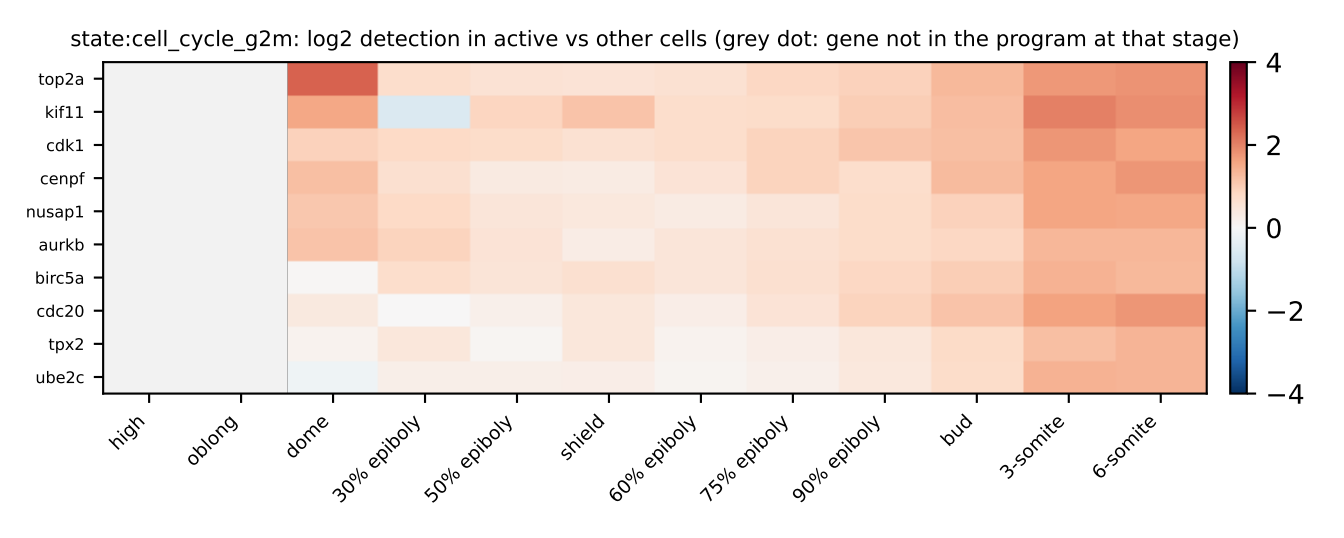

signal:BMP: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
bmp7a
bmp2b
bambia
id1
id3
msx1b
szl
bmp4
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
2
0
−2
−4

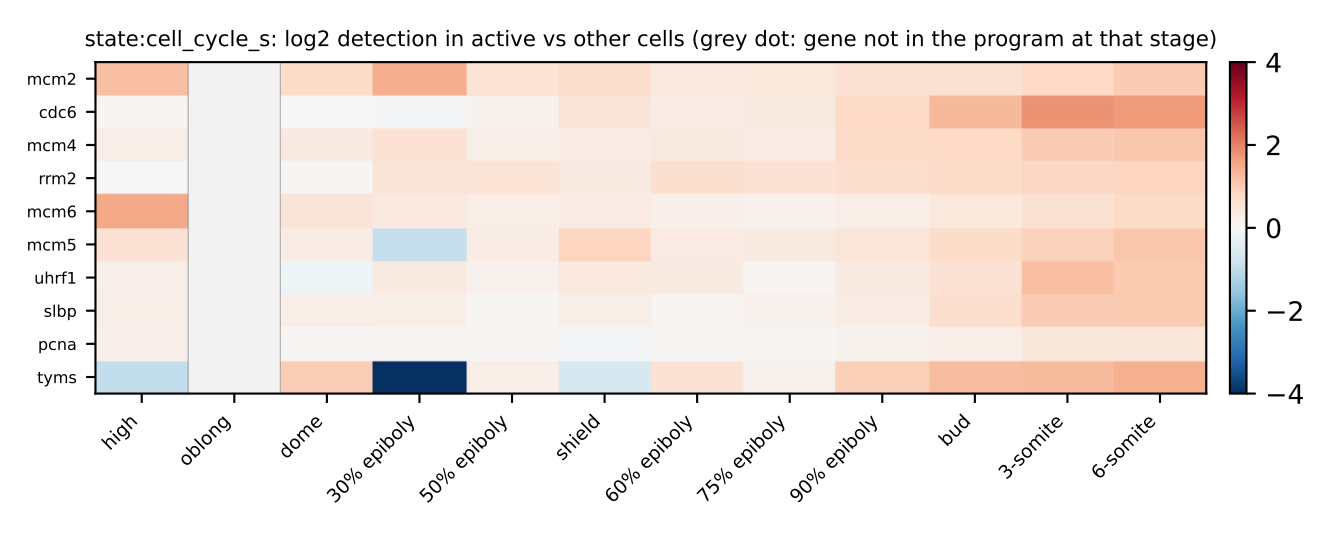

signal:FGF: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
fgf8a
il17rd
etv4
fgf3
fgf24
fgf4
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
2
0
−2
−4

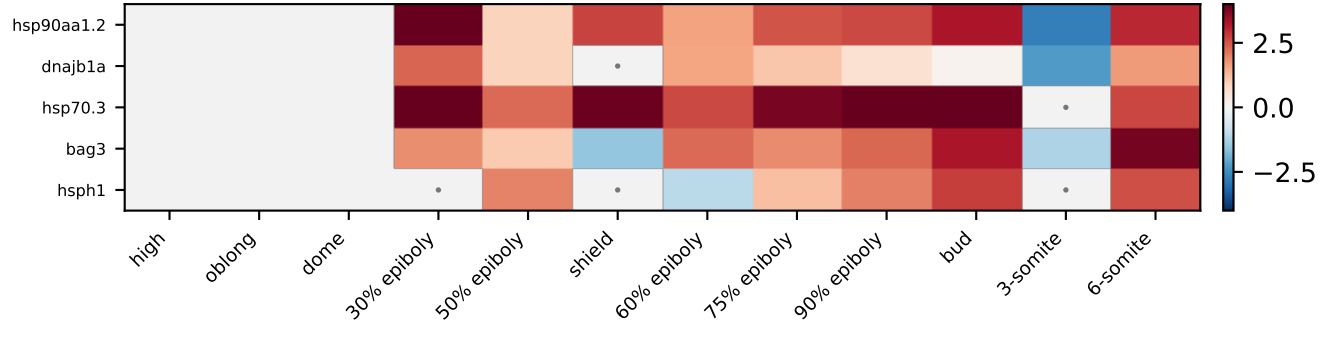

signal:Nodal: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
lft2
lft1
ndr1
ndr2
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
2.5
0.0
−2.5

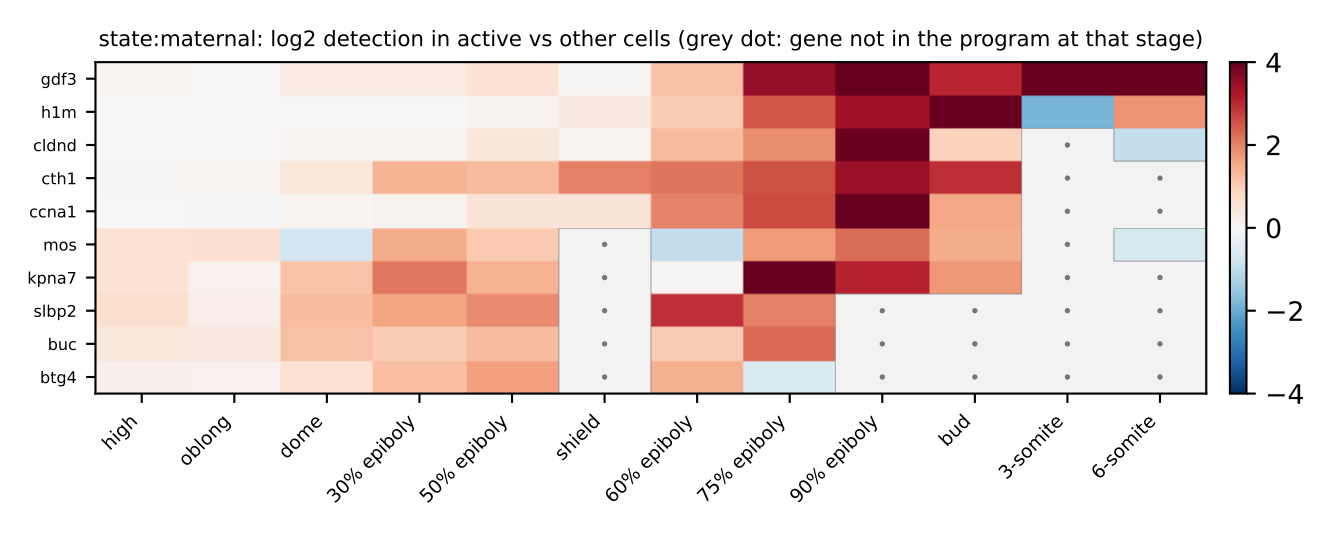

signal:Wnt: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
wnt8a
lef1
sp5l
sp5a
notum1a
wnt3a
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
2
0
−2
−4

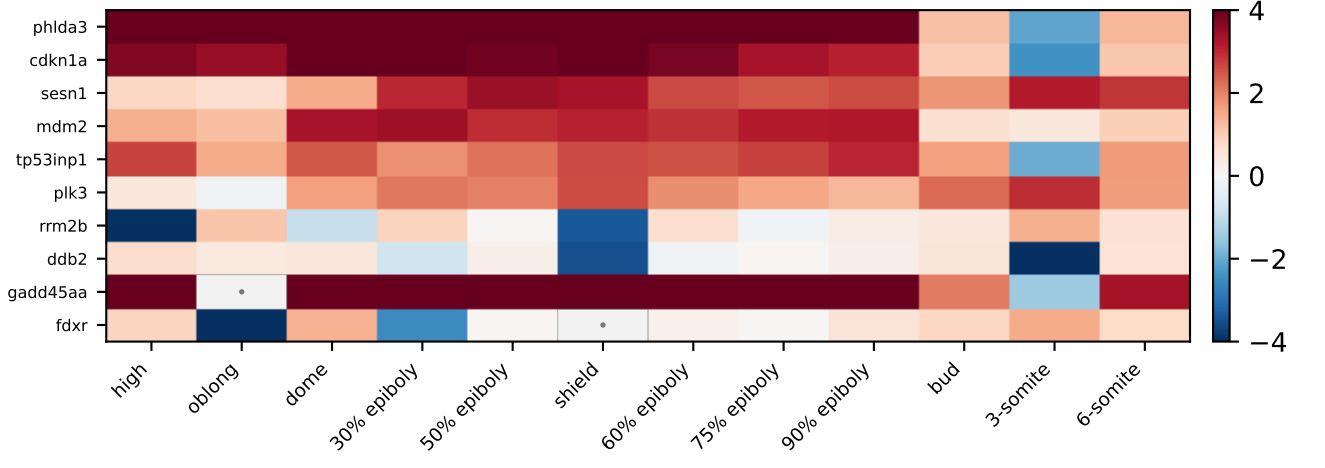

state:blastomere: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
sox19b
sox19a
sox3
pou5f3
lin28a
nanog
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
2
0
−2
−4


state:cell_cycle_g2m: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
top2a
kif11
cdk1
cenpf
nusap1
aurkb
birc5a
cdc20
tpx2
ube2c
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
2
0
−2
−4


state:cell_cycle_s: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
mcm2
cdc6
mcm4
rrm2
mcm6
mcm5
uhrf1
slbp
pcna
tyms
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
2
0
−2
−4


state:heat_shock: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
hsp90aa1.2
dnajb1a
hsp70.3
bag3
hsph1
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
2.5
0.0
−2.5


state:maternal: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
gdf3
h1m
cldnd
cth1
ccna1
mos
kpna7
slbp2
buc
btg4
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
2
0
−2
−4


state:p53_response: log2 detection in active vs other cells (grey dot: gene not in the program at that stage)
phlda3
cdkn1a
sesn1
mdm2
tp53inp1
plk3
rrm2b
ddb2
gadd45aa
fdxr
high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite
4
2
0
−2
−4